

★ WEEK 12 · PART 1 of 3

Interest-Rate Models: The Short Rate

Vasicek and CIR — Last Week's OU Machinery, Applied to Rates

What makes rates different · Vasicek · negative rates · CIR

Rates revert, they don't wander. We apply the Ornstein-Uhlenbeck process to the most important quantity in fixed income — the short rate — and meet the two classic models built on it.

1

PART 1 · THE SHORT RATE

Vasicek, CIR, and Mean-Reverting Rates

What sets interest rates apart, the short-rate modeling idea, the Vasicek model as OU for the rate, its negative-rate catch, and the CIR square-root diffusion that fixes it.

When Rates Went Wild — Part 1

Why a stock model can't price a bond

You cannot price a bond with a stock model. A stock can drift up forever; an interest rate cannot. Rates are pulled back toward a long-run level, and every bond's value depends on the whole future path of that rate.

Why stocks and rates differ

- A stock price can trend away without limit.
- A rate is tethered — it reverts to a normal level.
- GBM captures the first; it cannot capture the second.

Why it matters for bonds

- A bond's price is the discounted future short rate.
- So you must model how the rate EVOLVES, not just today's.
- That needs a mean-reverting model of the rate.

When Rates Went Wild — Part 2

Vasicek, CIR, and the lesson

In 1977 Oldrich Vasicek modeled the short rate as an OU process — mean-reverting, but able to go negative. In 1985 Cox, Ingersoll and Ross added a square-root diffusion to keep it positive. The lesson: the diffusion matters as much as the drift.

The two models

- Vasicek: OU applied directly to the rate.
- CIR: the same drift, a square-root diffusion.
- Both are mean-reverting; they differ in the noise term.

The lesson

- Choosing the drift is only half the model.
- The diffusion decides whether rates can go negative.
- Small SDE choices have large financial consequences.

From OU to Interest Rates

Where Week 11 leads

1

We built the OU process

A mean-reverting SDE: a spring pulling a state back to a long-run mean, with three parameters and a half-life.

2

Interest rates are the natural application

The short rate reverts to a normal level rather than wandering. It is exactly what OU was built to describe.

3

Two classic short-rate models follow

Vasicek is OU for the rate; CIR modifies the diffusion so rates stay positive. This week turns OU into fixed-income machinery.

Part 1 in one line: apply the OU process to the short rate to get Vasicek and CIR.

What Makes Interest Rates Different

Not just another asset price

Interest rates behave unlike stock prices. They revert to a normal level rather than trending; they can't grow without bound; and a single rate today implies a whole curve of yields across maturities. Modeling them needs its own toolkit.

How rates behave

- They revert toward a long-run, policy-anchored level.
- They don't compound upward like a stock.
- They move within a bounded, economically-set range.

What that implies

- A mean-reverting model, not GBM.
- One rate must generate yields for all maturities.
- The whole future path of the rate matters.

The Short-Rate Modeling Idea

One rate to generate them all

The setup

- Model a single instantaneous 'short rate'.
- Everything else is derived from its dynamics.
- Bond prices come from discounting at this rate.

Why it is powerful

- One SDE implies an entire yield curve.
- A few parameters describe the whole term structure.
- It is the foundation of classical rate modeling.

The Vasicek Model

The OU process, applied to the short rate

$$\text{Vasicek: } dr_t = \kappa(\theta - r_t) dt + \sigma dW_t$$

What it says

- The rate is pulled toward a long-run level theta.
- The pull has strength kappa — the reversion speed.
- A constant-size Brownian shock adds randomness.

Why it works

- It is literally the OU process, now for the rate.
- Mean reversion matches how rates actually behave.
- It yields closed-form bond prices — very tractable.

Vasicek Is OU for the Short Rate

Nothing new — just a new application

The mapping

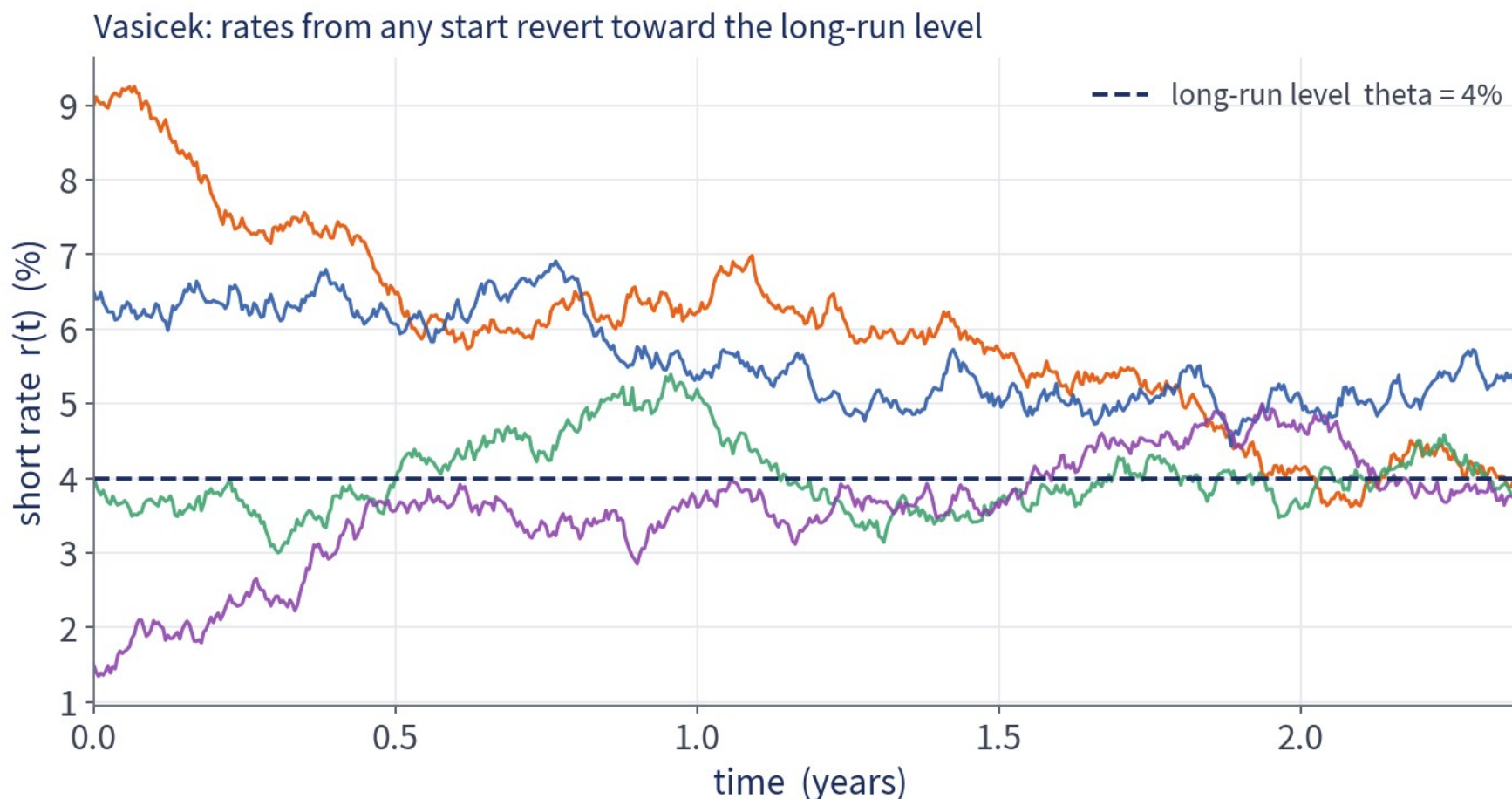
- The state X becomes the short rate r .
- θ is the long-run rate; κ the reversion speed.
- σ is the rate's volatility.

What carries over

- The half-life, $\log_2$ over κ , still applies.
- The stationary distribution is still Gaussian.
- Estimation still runs through the AR(1) bridge.

Vasicek = OU for the Short Rate

Rates starting high, low, or on target are all pulled toward the long-run level θ . The same mean-reverting spring from Week 11, now describing the most important rate in finance.



Vasicek's Mean Reversion

Rates pulled home to a long-run level

The long-run level

- Theta is the level rates revert toward.
- It reflects the normal, policy-anchored rate.
- Deviations are temporary, not permanent.

The speed of return

- Kappa sets how fast rates come back.
- The half-life is $\log_2$ over kappa.
- Faster reversion means shorter-lived swings.

Vasicek's Catch: Negative Rates

Constant volatility can push rates below zero

Vasicek's shock has a constant size regardless of the rate's level. So even when the rate is near zero, a big enough downward shock can push it below zero — something long seen as a flaw, though real rates did turn negative in the 2010s.

Why it happens

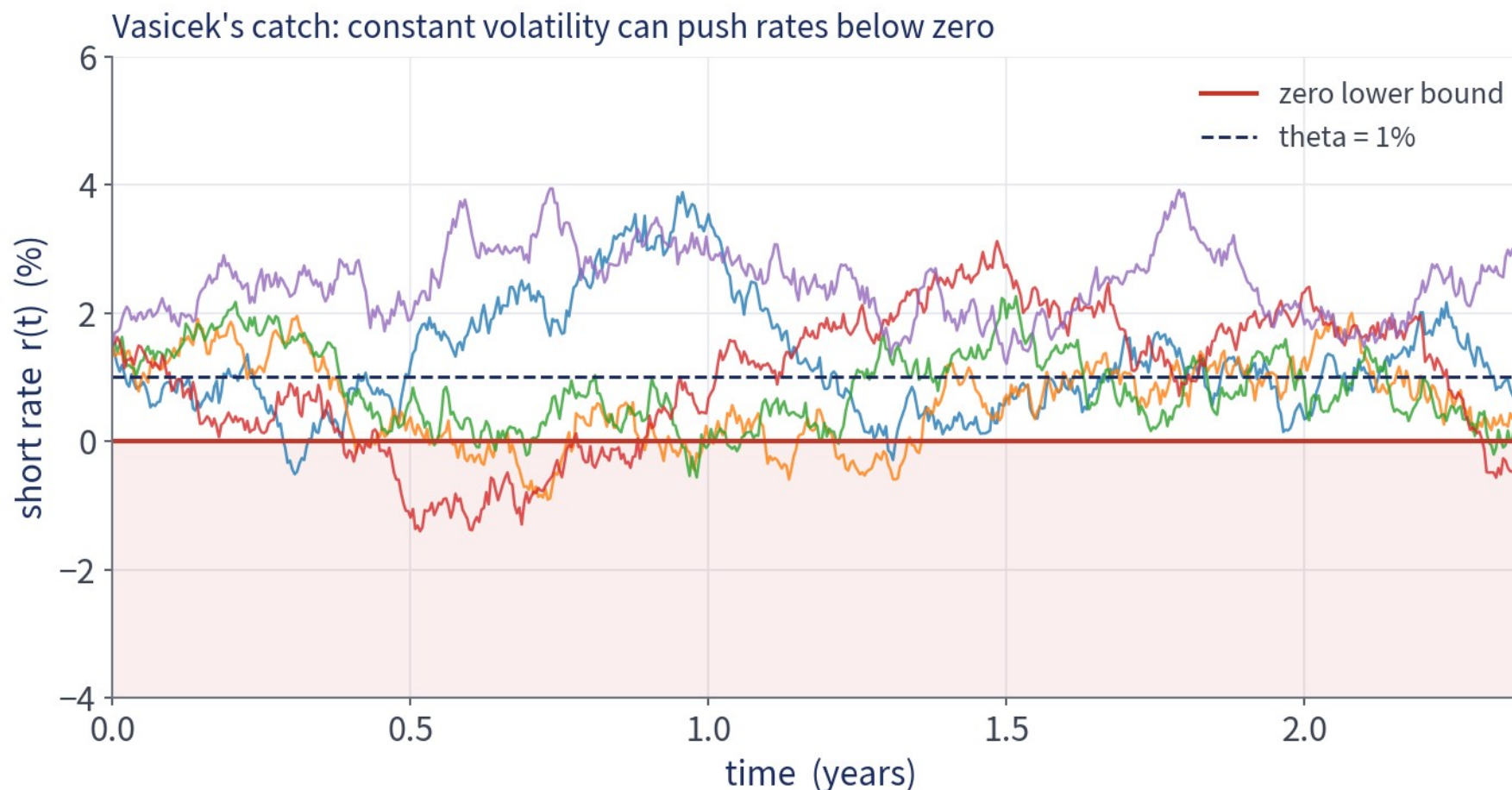
- The diffusion is constant — it doesn't shrink near zero.
- A large negative shock isn't damped at low rates.
- The Gaussian distribution has support below zero.

How much it matters

- For decades it was treated as an obvious defect.
- But some economies did see negative policy rates.
- Still, many applications want a strictly positive rate.

The Negative-Rate Problem, Visualized

With a low long-run level and constant volatility, several Vasicek paths dip below zero (shaded). The constant diffusion offers no floor — the model has no built-in respect for a zero bound.



The CIR Model

A square-root diffusion keeps rates positive

$$\text{CIR: } dr_t = \kappa(\theta - r_t) dt + \sigma \sqrt{r_t} dW_t$$

What changed

- The drift is identical to Vasicek — same mean reversion.
- The diffusion is now sigma times the square root of the rate.
- So volatility shrinks as the rate approaches zero.

The effect

- Near zero, shocks become tiny and can't push through.
- The rate is repelled from the zero boundary.
- It stays non-negative, under a mild condition.

The Square-Root Diffusion

Why the noise term is the whole story

diffusion:

$\underbrace{\sigma}_{\text{Vasicek: constant}}$

VS

$\underbrace{\sigma\sqrt{r_t}}_{\text{CIR: shrinks as } r \rightarrow 0}$

The contrast

- Vasicek: a constant diffusion at every rate level.
- CIR: a diffusion that scales with the root of the rate.
- Same drift, entirely different behavior near zero.

Why it helps

- As the rate falls to zero, its volatility vanishes.
- So the rate cannot be knocked negative.
- Volatility also rises with the level, matching data.

Why CIR Stays Positive

The Feller condition

Feller condition: $2\kappa\theta \geq \sigma^2 \Rightarrow r_t > 0 \text{ always}$

What it says

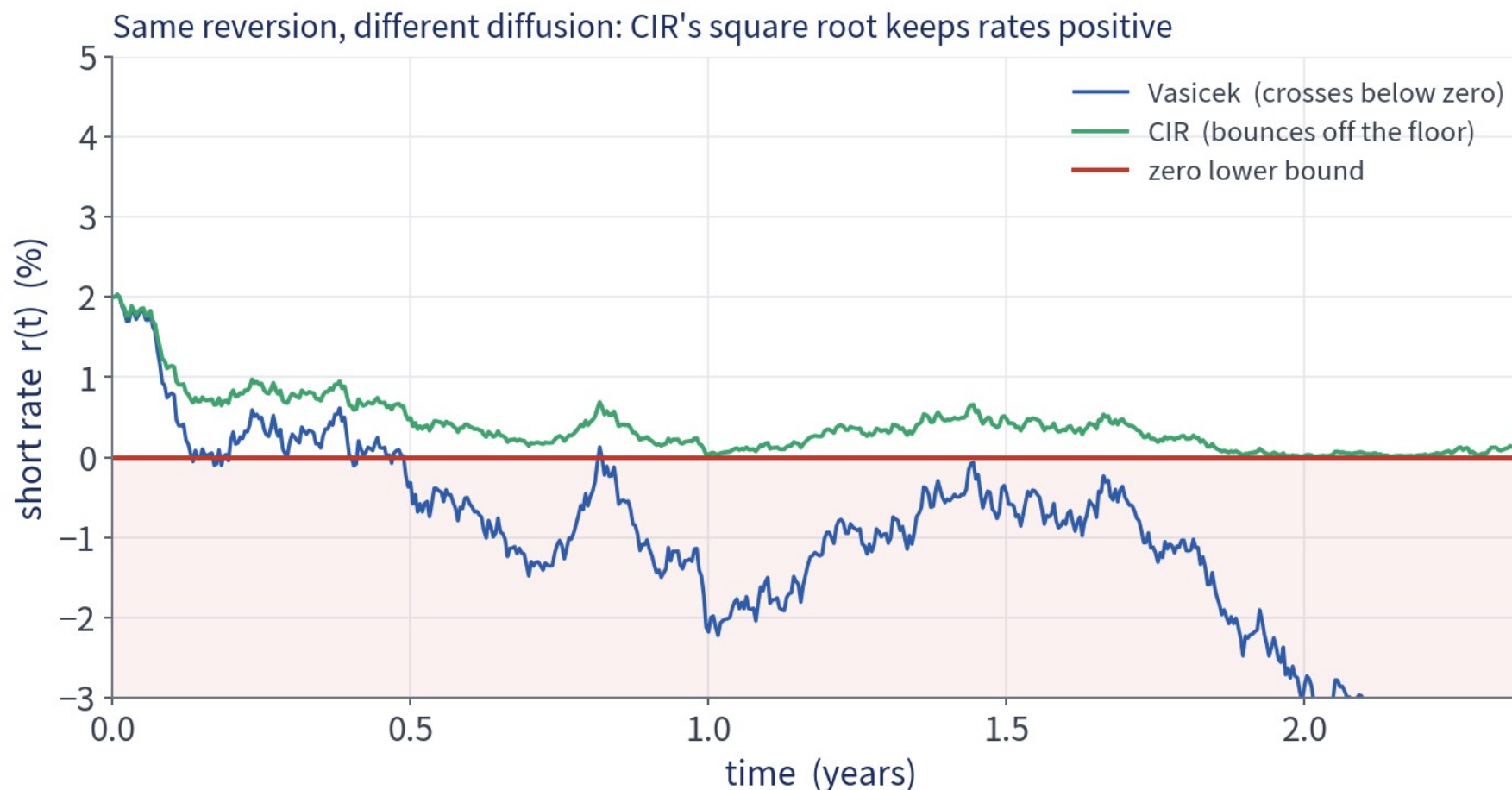
- If reversion and mean are strong enough relative to noise ...
- ... the rate is guaranteed to stay strictly positive.
- Formally, twice kappa times theta at least sigma squared.

Reading it

- Strong pull and a high mean keep the rate up.
- Large volatility fights against positivity.
- Meet the condition and zero is never reached.

CIR Bounces Off the Floor

From the same low start, Vasicek (blue) crosses below zero while CIR (green) is repelled by its vanishing volatility. Same reversion, different diffusion — and a very different result at the boundary.



CIR vs Vasicek at a Glance

Same reversion, different diffusion

Feature	Vasicek	CIR
Drift	$\kappa(\theta - r)$ — reverting	same — reverting
Diffusion	σ (constant)	$\sigma \sqrt{r}$ (level-dependent)
Negative rates	possible	ruled out (Feller)
Distribution	Gaussian	non-central chi-squared
Bond prices	closed-form	closed-form, harder

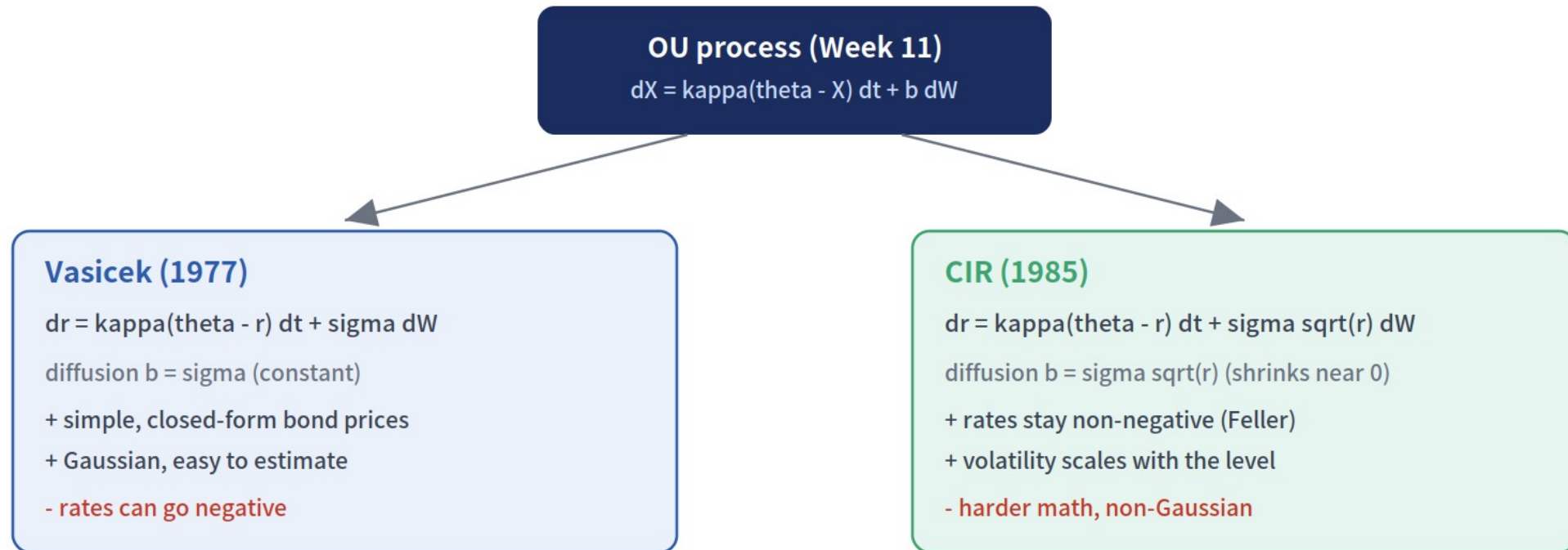
Both share the mean-reverting drift; the diffusion is the single design choice that sets them apart.

Short-Rate Models Are OU Cousins

Both descend from the OU process with the same mean-reverting drift. Vasicek keeps a constant diffusion; CIR uses a square-root diffusion that vanishes at zero. The drift is shared; the noise is the choice.

Short-Rate Models Are OU Cousins

Same mean-reverting drift as last week - the models differ only in their diffusion.



Same drift, different diffusion: the choice of $b(r)$ is what separates the two classic short-rate models.

When to Use Which

Trade-offs between the two

Reach for Vasicek

- You want the simplest, most tractable model.
- Gaussian rates and easy estimation are fine.
- The zero bound isn't a practical concern here.

Reach for CIR

- You need rates to stay strictly positive.
- You want volatility to scale with the level.
- You can accept somewhat harder mathematics.

Both Are OU Cousins

One idea, two diffusions

What they share

- The same mean-reverting drift, κ times the gap.
- A long-run level θ and a reversion speed κ .
- The whole OU intuition from Week 11.

What differs

- Only the diffusion term: constant vs square-root.
- That one choice decides the zero-bound behavior.
- Everything else is the OU machinery you know.

What Parts 2 and 3 Add

From the short rate to prices and code

Part 2 — the term structure

- How one short rate implies a whole yield curve.
- Discounting the random future rate into bond prices.
- Normal, inverted, and humped curve shapes.

Part 3 — computation

- Discretizing these SDEs (Euler and Milstein).
- Calibrating the models to market data.
- Pricing a bond end-to-end by simulation.

Simulate Vasicek and CIR

Direct Claude Code to see the zero floor

GOAL Simulate both short-rate models and watch how CIR's square-root diffusion keeps rates positive.

[INPUT] Prompt · Claude Code

Write and run `short_rates.py` that:

1. simulates Vasicek $dr = \kappa(\theta - r)dt + \sigma dW$ and CIR $dr = \kappa(\theta - r)dt + \sigma \sqrt{r} dW$ (Euler steps);
2. uses a low θ and high σ so Vasicek can go negative; plots several paths of each;
3. counts how often each model goes below zero;
4. checks the CIR Feller condition $2\kappa\theta \geq \sigma^2$ and reports whether it holds.

Show Vasicek dipping below zero while CIR does not.

How to work with it

- Both share the same reverting drift.
- Only the diffusion term differs.
- CIR's vol vanishes as r approaches zero.
- Feller holding means rates stay positive.

Deliverable: `short_rates.py` + Vasicek and CIR paths, a zero-crossing count, and a Feller check.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1 What process is the Vasicek model, applied to the short rate?

The OU process

Q2 Why can Vasicek rates go negative?

Constant diffusion

Q3 How does CIR keep rates non-negative?

Square-root diffusion

All three come from the OU link, the constant diffusion, and the CIR square-root term.

Three Misconceptions to Avoid

Where rate-model intuition slips

Common myths

- "You can model rates like a stock with GBM."
- "Vasicek and CIR have different mean reversion."
- "CIR can never, under any parameters, hit zero."

The reality

- Rates revert; GBM trends — use a mean-reverting SDE.
- Their DRIFT is identical; only the diffusion differs.
- CIR stays positive only if the Feller condition holds.

Three Things to Carry Forward

Part 1 in one screen

1

Rates revert — so model the short rate

Interest rates are pulled toward a long-run level rather than trending. The short-rate approach models one instantaneous rate, and everything else follows from its dynamics.

2

Vasicek is OU for the rate

The same mean-reverting SDE from Week 11, now describing the short rate. Tractable and Gaussian — but its constant diffusion lets rates fall below zero.

3

CIR fixes the floor with a square root

Keeping the reverting drift but scaling volatility by the root of the rate, CIR makes shocks vanish near zero. Under the Feller condition, rates stay strictly positive.

Next — Part 2: from the short rate to bond prices and the shapes of the yield curve.

★ WEEK 12 · PART 1 · PRACTICE

Short-Rate Models, Worked by Hand

Vasicek's Negative-Rate Catch, and CIR's Positive Floor

OU for interest rates, and the square-root fix

Part 1's short-rate models, with the arithmetic shown. We compute the probability Vasicek assigns to negative rates, then check CIR's Feller condition and see how its square-root diffusion vanishes at zero to keep rates positive.

1P

PART 1 · APPLICATIONS

Two Rate Models, by Hand

Case A: Vasicek as OU for the short rate - its stationary distribution, and the probability it goes negative. Case B: CIR's square-root diffusion and the Feller condition that keeps rates strictly positive. Then a Claude Code lab.

What Part 1 Told Us

Now we make it concrete - with numbers

1

Vasicek is OU for the short rate

The same mean-reversion SDE from Week 11 - rates pulled home to a long-run level θ at speed κ .

2

Vasicek's catch is negative rates

Its constant volatility makes the stationary distribution Gaussian, so it puts real probability below zero.

3

CIR fixes it with a square root

Scaling the noise by $\sqrt{r}$ makes it vanish at zero. The Feller condition guarantees rates stay strictly positive.

Everything here is a calculator exercise: one stationary distribution, one Feller check.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a rate-model SDE.
- Compute a probability or a condition by hand.
- Then read what it means for rates.

What you should watch

- How constant vol lets rates go negative.
- How the Feller condition guards positivity.
- How the square-root noise dies at zero.

Vasicek's Negative-Rate Catch

A Gaussian tail below zero

Vasicek is just the OU process applied to the short rate. That makes its long-run distribution a Gaussian centered on the mean-reversion level - which means it assigns real probability to rates below zero. With a modest mean and a typical volatility, we compute exactly how much

The parameters

- Reversion speed $\kappa = 0.30$.
- Long-run level $\theta = 2.5\%$.
- Volatility $\sigma = 1.5\%$.

What we'll do by hand

- Find the stationary mean and sd.
- Standardize zero into a z-score.
- Read off the probability of a negative rate.

Vasicek and Its Stationary Law

OU, with a Gaussian equilibrium

$$\begin{aligned} \text{Vasicek: } dr &= \kappa(\theta - r) dt + \sigma dW && (\text{OU, applied to the short rate}) \\ \text{stationary: } r_\infty &\sim N\left(\theta, \frac{\sigma^2}{2\kappa}\right) && (\text{constant } \sigma \Rightarrow \text{Gaussian}) \end{aligned}$$

The dynamics

- Identical to OU - a pull toward theta.
- The diffusion is a constant sigma.
- That constant is the source of the problem.

The equilibrium

- The stationary law is a normal distribution.
- Centered at theta, with a fixed spread.
- And a normal has tails on both sides.

The Stationary Mean and SD

Where rates settle, and how wide

$$\text{mean} = \theta = 2.5\%, \quad \text{sd} = \frac{\sigma}{\sqrt{2\kappa}} = \frac{1.5\%}{\sqrt{0.60}} = \mathbf{1.94\%}$$

The mean

- Rates settle around $\theta = 2.5\%$.
- That's the long-run reversion level.
- The center of the distribution.

The spread

- The sd is sigma over root two-kappa.
- That works out to 1.94%.
- Wide enough to reach below zero.

The Probability of Negative Rates

Standardize zero, and read the tail

$$P(r < 0) = \Phi\left(\frac{0-2.5\%}{1.94\%}\right) = \Phi(-1.29) = \mathbf{9.8\%} \quad (\text{Vasicek's negative-rate catch})$$

The calculation

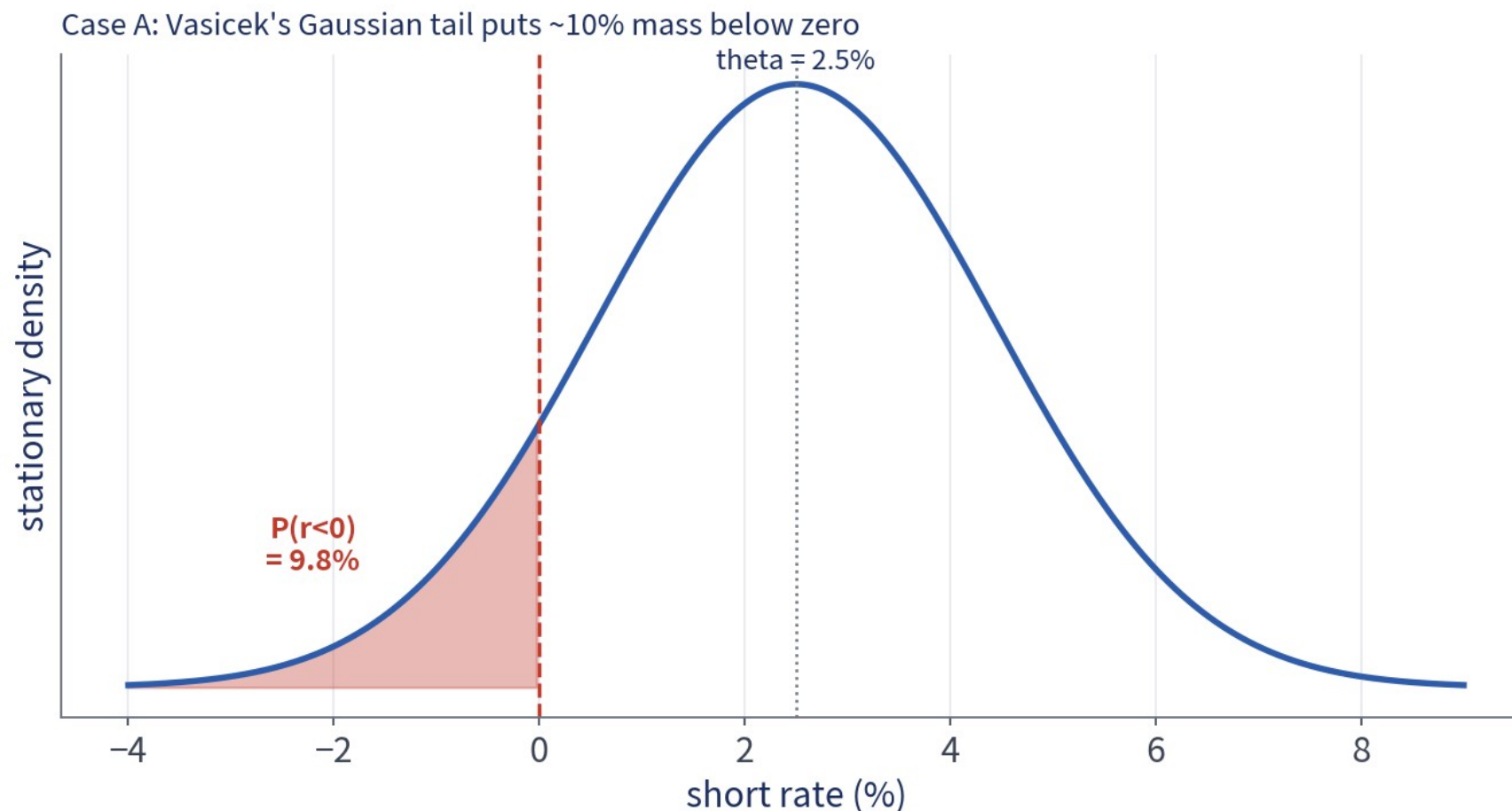
- Zero is 1.29 sd below the 2.5% mean.
- The normal CDF at minus-1.29 is 0.098.
- So about a 10% chance of negative rates.

The verdict

- That's not negligible - it's one year in ten.
- The constant-vol assumption did this.
- This is Vasicek's structural weakness.

A Tenth of the Mass Below Zero

The stationary distribution is a bell curve centered at 2.5%, but its left tail crosses zero. The shaded region - about 10% of the probability - is where Vasicek says rates are negative. Historically awkward, and the direct cost of a constant volatility



Why the Catch Matters

Constant vol has a price

The problem

- A Gaussian has no floor.
- The lower the mean, the more mass below zero.
- Bond prices from it can misbehave.

When it's tolerable

- Negative rates are no longer purely theoretical.
- Vasicek stays analytically clean and tractable.
- For many uses the 10% tail is acceptable.

CIR and the Feller Condition

A square root keeps rates positive

CIR keeps the same mean reversion but scales the noise by the square root of the rate. As the rate falls toward zero, the volatility shrinks with it, so the randomness can't push it across the boundary. The Feller condition tells you exactly when this guarantee holds. We check it by hand

The model

- Same reversion: $\kappa = 0.30$, $\theta = 3\%$.
- But the diffusion is σ times $\sqrt{r}$.
- Two candidate volatilities to test.

What we'll do by hand

- Compute $2\kappa\theta$.
- Compare it to σ^2 for each.
- See the diffusion vanish as r goes to zero.

CIR and Feller

Where the square root earns its keep

CIR: $dr = \kappa(\theta - r) dt + \sigma\sqrt{r} dW$ (square-root diffusion)

Feller: $2\kappa\theta \geq \sigma^2 \Rightarrow$ the rate stays strictly positive

The diffusion

- The noise is sigma times root-r.
- It shrinks as the rate approaches zero.
- That's what protects the boundary.

The condition

- Feller: two-kappa-theta at least sigma-squared.
- Then the rate never touches zero.
- Reversion must beat the noise.

Checking the Feller Condition

Two volatilities, two verdicts

$$2\kappa\theta = 2(0.30)(0.03) = 0.018$$

$$\sigma = 0.10 : \sigma^2 = 0.010 \leq 0.018 \Rightarrow \text{HOLDS (r \> 0)}$$

$$\sigma = 0.15 : \sigma^2 = 0.0225 > 0.018 \Rightarrow \text{VIOLATED (r can hit 0)}$$

With two-kappa-theta at 0.018, a volatility of 0.10 satisfies Feller and rates stay strictly positive; 0.15 violates it, and the rate can reach zero.

The Noise Dies at the Floor

Why zero is a wall, not a door

$$\sigma\sqrt{r} : \quad r = 3\% \rightarrow 0.017, \quad r = 0.5\% \rightarrow 0.007, \quad r \rightarrow 0 \Rightarrow 0 \quad (\text{the noise dies at the floor})$$

The mechanism

- At 3%, the diffusion is a healthy 0.017.
- At 0.5%, it's already halved to 0.007.
- At zero, it's exactly zero.

The consequence

- With no noise at the boundary, only drift acts.
- And drift points up, back toward theta.
- So the rate is pushed away from zero.

One Dips Below, One Bounces Off

Driven by the same shocks, the Vasicek path (red) plunges below zero when its constant volatility overwhelms the weak upward drift, while the CIR path (green) has its noise fade near the floor and bounces off zero. Same reversion, different diffusion, opposite behavior at the boundary.

Case B: constant vol lets rates go negative; sqrt diffusion holds the floor



Why CIR for Rates

Positivity, built into the diffusion

What CIR buys

- Rates that respect a zero floor (under Feller).
- Volatility that rises with the level - realistic.
- Still tractable, with closed-form bond prices.

The trade-off

- The math is a bit heavier than Vasicek.
- The Feller condition can bind on calibration.
- But no embarrassing negative rates.

Same Reversion, Different Diffusion

The one line that separates them

Vasicek and CIR share the exact same mean-reversion drift; they differ only in the noise term. Vasicek's constant sigma gives a clean Gaussian - and a negative tail. CIR's sigma-times-root-r gives a floor at zero - at the cost of a Feller condition. The whole choice lives in the diffusion

Vasicek

- Constant volatility.
- Gaussian, negative tail (10% here).
- Simplest, cleanest, but can go below zero.

CIR

- Volatility scales with root-r.
- Positive under Feller ($2\kappa\theta \geq \sigma^2$).
- Realistic and positive, slightly heavier.

Reproduce the Tail and the Floor

Direct Claude Code to check your arithmetic

GOAL Verify Vasicek's negative-rate probability and CIR's Feller check, then simulate both.

[INPUT] Prompt · Claude Code

```
Write and run part1_practice.py that:
CASE A - Vasicek:
  * from kappa=0.30, theta=0.025, sigma=0.015,
    print the stationary sd = sigma/sqrt(2 kappa)
    and P(r<0) via the normal CDF.
CASE B - CIR:
  * with kappa=0.30, theta=0.03, print 2 kappa theta
    and check Feller for sigma = 0.10 and 0.15;
  * print sigma*sqrt(r) at r = 3%, 0.5%, 0.
Then simulate Vasicek and CIR paths and count
how often each goes negative.
Match every printed number to the slides.
```

Check against the slides

- stationary sd = 1.94%.
- $P(r < 0) = 9.8\%$.
- $2 \text{ kappa theta} = 0.018$.
- Feller: 0.10 holds, 0.15 fails.

Deliverable: part1_practice.py whose printed numbers match the negative-rate tail and the Feller check.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Why can Vasicek produce negative rates?

Constant vol makes the law Gaussian

Q2 What is the Feller condition?

2-kappa-theta at least sigma-squared - rates stay positive

Q3 How does CIR keep rates positive?

The root-r diffusion vanishes at zero

All three come from Vasicek's stationary law and CIR's Feller condition.

Three Pitfalls to Avoid

Where rate models trip people

Common mistakes

- "Vasicek can't give negative rates."
- "CIR is always positive."
- "CIR and Vasicek differ in their drift."

The reality

- Its Gaussian tail puts real mass below zero - 10% here.
- Only if Feller holds; violate it and zero is reachable.
- The drift is identical - only the diffusion differs.

Two Calculations to Remember

Part 1 applications in one screen

1

Vasicek assigns real mass to negative rates

As OU for the short rate, its stationary law is normal - mean 2.5%, sd 1.94%. Standardizing zero gives a z of -1.29 and a probability of 9.8% that rates are negative. That's the price of a constant volatility.

2

CIR's square root holds the floor

With two-kappa-theta = 0.018, a volatility of 0.10 satisfies Feller and rates stay strictly positive, while 0.15 violates it. The diffusion sigma-root- r falls from 0.017 at 3% to zero at the boundary - so the noise dies exactly where it would do harm.

3

Same reversion, different diffusion

Vasicek and CIR share the mean-reversion drift and differ only in the noise. Constant vol gives simplicity and a negative tail; root- r gives positivity and a Feller condition. The whole modeling choice is one term.

Next — Part 2 practice: from the short rate to the whole term structure of bond yields.

★ WEEK 12 · PART 2 of 3

The Term Structure and Bond Pricing

One Short Rate, a Yield for Every Maturity

Bond prices as expectations · the affine formula · yield-curve shapes

A single short-rate model implies an entire yield curve. We discount the random future rate into bond prices, turn prices into yields, and read the normal, inverted, and humped shapes the model can produce.

2

PART 2 · THE YIELD CURVE

From the Short Rate to the Curve

The term structure, bond prices as expectations of the discounted short rate, the Vasicek affine formula, yields from prices, and the shapes — normal, inverted, humped — one model can make.

From a Rate to a Whole Curve

Part 1 modeled the rate; Part 2 prices bonds

1

We have a short-rate model

Vasicek or CIR: a mean-reverting SDE for the instantaneous rate, with a few interpretable parameters.

2

But we quote yields, not the short rate

Markets trade bonds across maturities. We need the short rate to imply a yield for every term.

3

Discounting bridges the two

A bond's price is the expected discounted payoff at the random short rate. From prices come yields — and the whole curve.

Part 2 in one line: discount the random short rate into bond prices, then read off the yield curve.

The Term Structure of Rates

One short rate, a yield for every maturity

The term structure — the yield curve — is the set of yields across all maturities at one moment. A short-rate model must produce this entire curve from the single instantaneous rate and its dynamics.

What it is

- A yield for every maturity, from months to decades.
- Plotted, it is the familiar yield curve.
- It summarizes the price of money over time.

The modeling task

- Turn one short rate into yields at all terms.
- The dynamics of the rate shape the whole curve.
- A few parameters must generate the entire structure.

What a Yield Curve Is

Yields plotted against maturity

Reading it

- The horizontal axis is maturity — the term.
- The vertical axis is the yield for that term.
- Its slope and shape carry economic information.

Why it matters

- It prices every fixed-income instrument.
- Its slope reflects expectations and risk premia.
- Its inversions have famously preceded recessions.

From the Short Rate to Bond Prices

Discount the future at the random short rate

A zero-coupon bond pays one unit at maturity. Its price today is that payoff discounted along the future path of the short rate — but the rate is random, so the price is an EXPECTATION over all its possible paths.

The logic

- Discount the future payoff at the short rate as it evolves.
- The rate's path is uncertain, so average over paths.
- That average is the bond's fair price.

Why an expectation

- Each possible rate path gives a different discount.
- The price weighs them by their probabilities.
- This is where the model's dynamics enter.

The Bond Price as an Expectation

Averaging the discount over rate paths

$$P(0, T) = \mathbb{E} \left[\exp \left(- \int_0^T r_s ds \right) \right] \quad (\text{discount at the random short rate})$$

Reading it

- Inside: the discount factor over the whole path.
- The integral accumulates the short rate over time.
- The expectation averages across all possible paths.

What it needs

- The dynamics of the short rate — our SDE.
- For Vasicek and CIR, this has a closed form.
- So no simulation is needed for the price itself.

Why It Averages Over Paths

The rate is random, so the discount is too

The randomness

- The short rate follows a stochastic path.
- A high-rate path discounts the payoff heavily.
- A low-rate path discounts it lightly.

The resolution

- The price is the probability-weighted average.
- Mean reversion keeps the paths well-behaved.
- The model turns this average into a formula.

The Vasicek Bond Price

An affine formula in the short rate

$$\text{Vasicek: } P(t, T) = A(\tau) e^{-B(\tau) r_t}, \quad B(\tau) = \frac{1 - e^{-\kappa \tau}}{\kappa}, \quad \tau = T - t$$

The structure

- The price is a factor A times an exponential of the rate.
- The exponent is minus B(tau) times the current rate.
- A and B depend only on the maturity and parameters.

Why it is elegant

- No simulation — the expectation is solved exactly.
- The log-price is linear in the short rate (affine).
- CIR has a similar, if messier, closed form.

From Bond Prices to Yields

One short step to the curve

$$\text{yield from price: } y(0, T) = -\frac{\ln P(0, T)}{T}$$

The definition

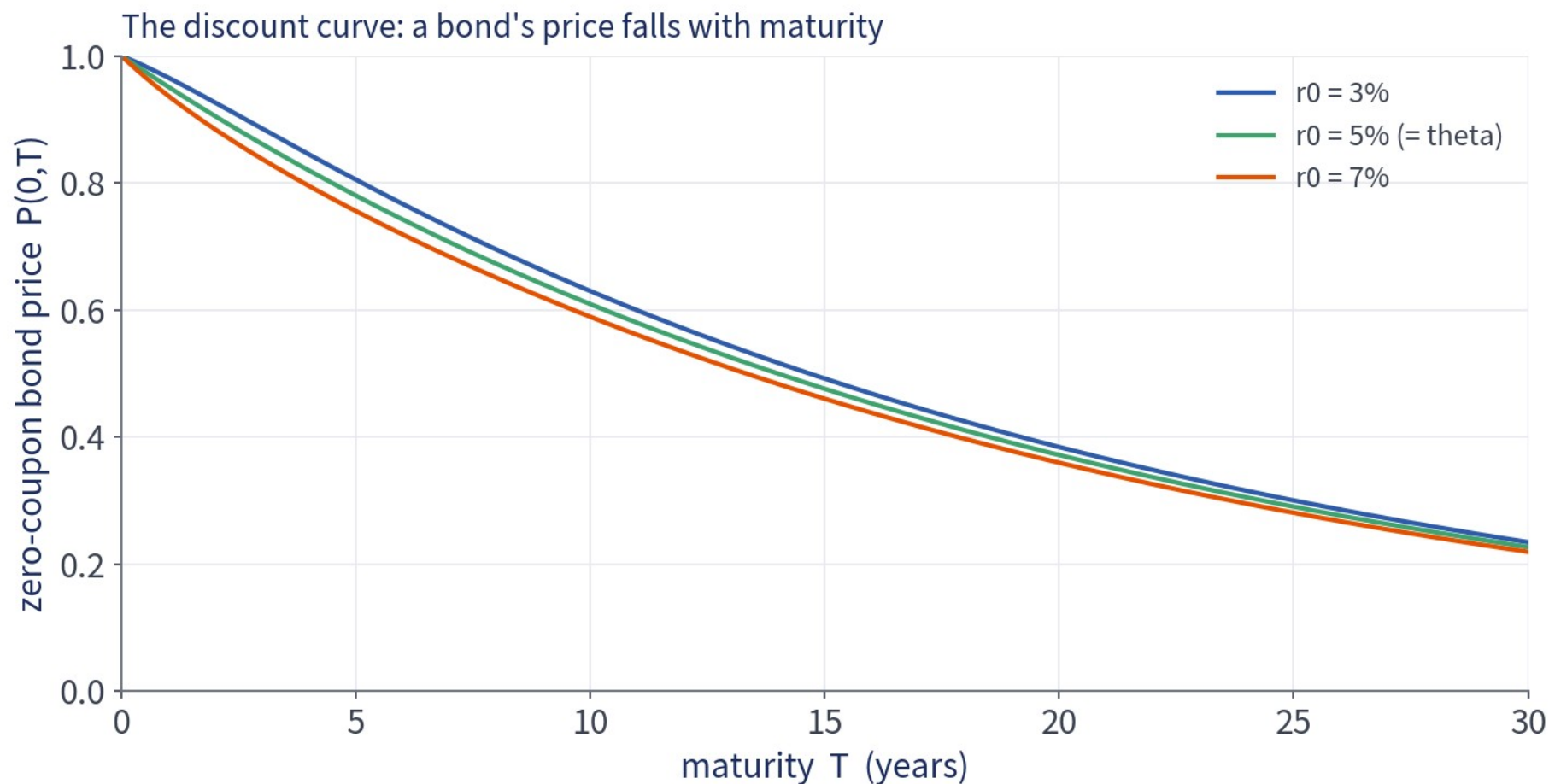
- The yield is minus the log-price divided by maturity.
- It is the constant rate that reproduces the price.
- Compute it at each maturity to trace the curve.

What you get

- A yield for every term — the whole curve.
- All from the one short rate and its parameters.
- Prices and yields are two views of the same thing.

The Discount Curve

The zero-coupon bond price falls with maturity: money further away is discounted more. A higher starting short rate discounts harder, pushing the whole price curve down.



Affine Term-Structure Models

Yields linear in the short rate

affine model: $y(0, T) = a(T) + b(T) r_0$ (the yield is linear in the short rate)

What 'affine' means

- Each yield is a linear function of the short rate.
- The intercept and slope depend on maturity.
- Vasicek and CIR are both affine models.

Why it is prized

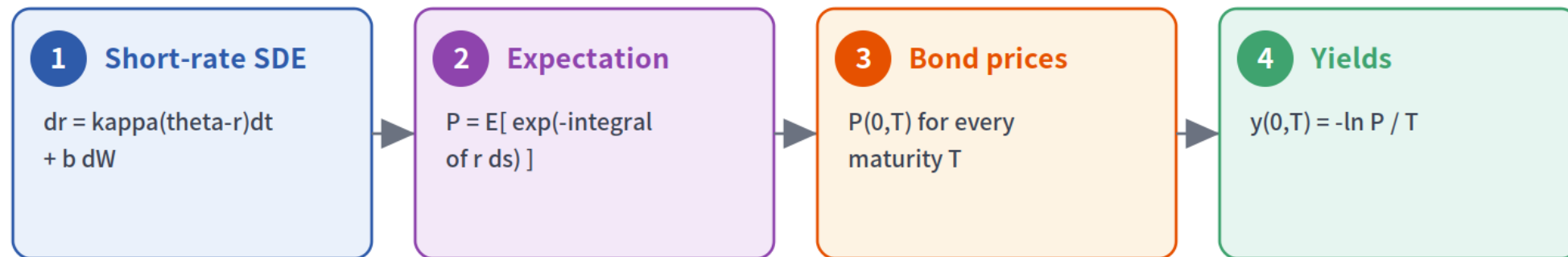
- Closed-form prices and yields — fast to compute.
- Easy to calibrate and to reason about.
- It is the workhorse class of rate models.

From One Short Rate to the Whole Curve

The short-rate SDE feeds an expectation of the discounted path, which gives bond prices at every maturity, which convert to yields. Fix the parameters and r_0 , and the model prints an entire curve.

From One Short Rate to the Whole Curve

The short-rate SDE feeds every step: expectation, then prices, then yields, then the curve.



The whole curve from a handful of parameters

Fix κ , θ , σ and the starting short rate r_0 , and the model prints a yield for every maturity - an entire term structure. In an AFFINE model (Vasicek, CIR), each yield is simply linear in r_0 .

One short rate today, a full curve of yields tomorrow.

What Shapes Can the Model Make?

Normal, inverted, and humped

A single short-rate model produces different curve shapes depending mainly on where the rate starts relative to its long-run level. From one model come upward, downward, and humped curves — the shapes seen in real markets.

The three basic shapes

- Normal: yields rise with maturity.
- Inverted: yields fall with maturity.
- Humped: yields peak at a middle maturity.

What drives them

- Mostly where today's rate sits versus the mean.
- Below the mean tends to give an upward curve.
- Above the mean tends to give an inverted one.

The Normal (Upward) Curve

The most common shape

When it appears

- The short rate is below its long-run level.
- The market expects rates to rise toward the mean.
- Longer bonds also carry a term premium.

How it reads

- Longer maturities yield more than shorter ones.
- It reflects a normal, expansionary environment.
- It is the shape most of the time.

The Inverted (Downward) Curve

The recession signal

When it appears

- The short rate is above its long-run level.
- The market expects rates to fall back down.
- Often tight policy today, easing expected ahead.

How it reads

- Longer maturities yield LESS than shorter ones.
- It has historically preceded recessions.
- A closely-watched macro warning sign.

The Humped Curve

A peak in the middle

What it is

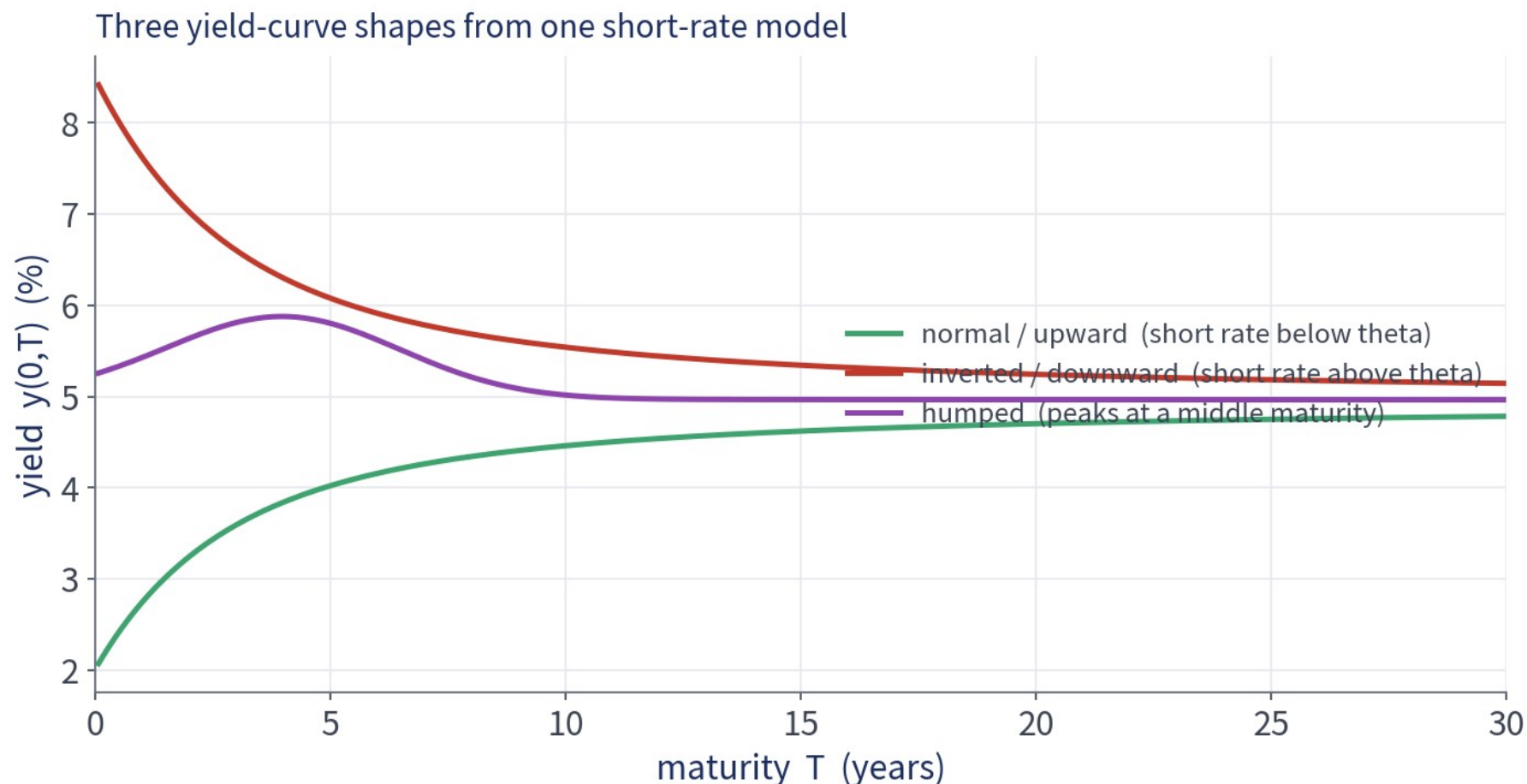
- Yields rise, peak at a medium term, then fall.
- A transitional shape between normal and inverted.
- It appears around turning points in policy.

Why the model makes it

- Mean reversion plus a term premium can peak mid-curve.
- Short and long yields converge to the long-run yield.
- Intermediate maturities can sit above both.

Yield-Curve Shapes

Normal (upward), inverted (downward), and humped curves — all produced by one short-rate model. The shape follows mainly from where today's short rate sits relative to its long-run level.



Today's Short Rate Sets the Shape

One parameter tilts the whole curve

Low starting rate

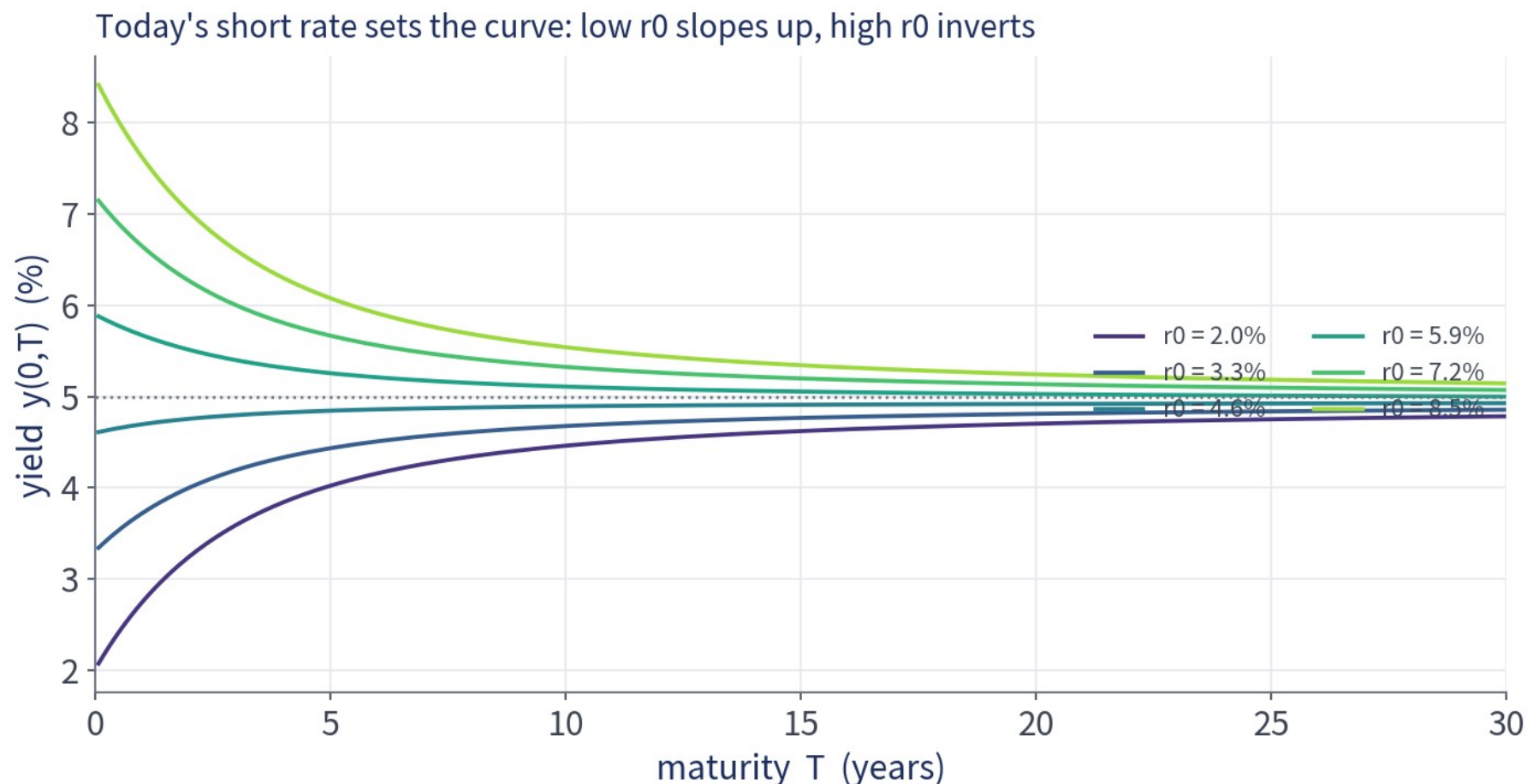
- Below the long-run level, rates are expected to climb.
- So the curve slopes upward.
- The classic normal shape.

High starting rate

- Above the long-run level, rates are expected to fall.
- So the curve slopes downward.
- The inverted shape.

The Short Rate Sets the Curve

Sweeping the starting short rate from low to high tilts the whole curve from upward-sloping to inverted. All curves converge to the same long-run yield — the reversion level shows through at the long end.



The Economic Reading

What the curve is telling you

The signal in the slope

- Upward: growth and gradually rising rates expected.
- Inverted: tight policy now, cuts expected ahead.
- The curve encodes the market's rate expectations.

A caution

- The model is a stylized lens, not the whole story.
- Real curves also reflect liquidity and supply effects.
- Read shapes as signals, not certainties.

What Part 3 Adds

From formulas to computation

Discretization

- Turn the SDEs into steps a computer can run.
- Euler-Maruyama, its pitfalls, and Milstein.
- The tools to simulate any short-rate model.

Calibration and pricing

- Fit the model's parameters to market data.
- Price a bond end-to-end by simulation.
- Cross-check simulation against the closed form.

Build a Yield Curve From a Model

Direct Claude Code to turn a rate into a curve

GOAL Use the Vasicek closed-form to build yield curves, and see how the starting short rate sets the shape.

[INPUT] Prompt · Claude Code

Write and run `yield_curve.py` that:

1. implements the Vasicek bond price
 $P(0,T) = A(T) \exp(-B(T) r_0)$ with
 $B(T) = (1 - e^{(-\kappa T)}) / \kappa$;
 2. converts prices to yields $y(T) = -\ln P / T$;
 3. plots the curve for several starting rates r_0 (below, at, and above θ);
 4. labels each curve normal, flat, or inverted.
- Show that low r_0 slopes up and high r_0 inverts, and that all curves meet at the long-run yield.

How to work with it

- The yield is linear in r_0 (affine).
- Low r_0 gives an upward curve.
- High r_0 gives an inverted curve.
- Long-maturity yields converge together.

Deliverable: `yield_curve.py` + Vasicek yields across maturities for several starting rates.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1 A zero-coupon bond price is the expectation of what?

The discounted short rate

Q2 In an affine model, how does a yield depend on the short rate?

Linearly

Q3 What mainly sets whether the curve is upward or inverted?

r_0 vs the long-run level

All three come from the pricing expectation, the affine form, and the starting short rate.

Three Misconceptions to Avoid

Where curve intuition slips

Common myths

- "You must simulate to price a Vasicek bond."
- "The curve shape is unrelated to today's rate."
- "An inverted curve guarantees a recession."

The reality

- Vasicek and CIR have closed-form bond prices.
- The starting short rate largely sets the shape.
- Inversion is a signal with a good record, not a certainty.

Three Things to Carry Forward

Part 2 in one screen

1

A bond price is an expectation

It is the expected discount of the payoff along the random path of the short rate. For Vasicek and CIR this expectation solves in closed form — no simulation needed to price.

2

Affine models give the whole curve

Each yield is linear in the short rate, so fixing the parameters and today's rate prints a yield for every maturity — an entire term structure from a handful of numbers.

3

One model, many shapes

Normal, inverted, and humped curves all come from a single short-rate model, driven mainly by where the rate starts relative to its long-run level.

Next — Part 3: discretizing the SDEs, calibrating to data, and pricing a bond by simulation.

★ WEEK 12 · PART 2 · PRACTICE

The Term Structure, Worked by Hand

Price a Bond, Then Build the Whole Curve

From the affine Vasicek formula to yields and shapes

Part 2's term structure, with the arithmetic shown. We price a zero-coupon bond with the affine Vasicek formula and turn it into a yield, then build the whole curve from today's short rate - and watch it flip from upward to inverted.

2P

PART 2 · APPLICATIONS

One Short Rate, a Whole Curve

Case A: price a 5-year zero with the affine Vasicek formula, then convert it to a yield. Case B: compute yields across maturities for a low and a high short rate, seeing the curve tilt from upward to inverted. Then a Claude Code lab.

What Part 2 Told Us

Now we price and build - with numbers

1

A bond price is an expectation

Discount the future at the random short rate and average over its paths. Vasicek makes that average a clean formula.

2

The Vasicek price is affine

Price equals A times e -to-the-minus- B -times- r . Take a log and divide by maturity, and you have the yield.

3

Today's short rate sets the shape

One number tilts the whole curve - upward when rates are low, inverted when they're high. We compute both.

Everything here is a calculator exercise: one bond price, one yield, two curve shapes.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from the affine Vasicek formula.
- Compute a price, a yield, or a curve.
- Then read the shape it produces.

What you should watch

- How the affine formula turns r into a price.
- How price becomes yield in one step.
- How the short rate flips the curve's slope.

Price a Vasicek Bond

From the short rate to a yield

Under Vasicek, a zero-coupon bond has an affine price: a maturity-dependent factor A times an exponential in today's short rate, with a coefficient B . We compute B and A for a five-year bond, price it at a 4% short rate, and turn that price into a yield.

The setup

- Vasicek: $\kappa = 0.30$, $\theta = 5\%$, $\sigma = 2\%$.
- Today's short rate is 4%.
- Maturity is five years.

What we'll do by hand

- Compute $B(5)$ and $A(5)$.
- Price the bond as A times e -to-the-minus- $B \cdot r$.
- Convert the price to a yield.

The Affine Bond Price

Price, linear in the short rate's exponent

$$P(t, T) = A(t, T) e^{-B(t, T) r_t}, \quad B(t, T) = \frac{1 - e^{-\kappa(T-t)}}{\kappa} \quad (\text{price is affine in the short rate})$$

The B factor

- It measures sensitivity to the short rate.
- It grows with maturity, then levels off.
- It's the duration-like term.

The A factor

- It captures the drift and volatility terms.
- It depends only on maturity, not on r .
- Together they give the full price.

Compute B and A

The two affine factors

$$B(5) = \frac{1 - e^{-0.30 \times 5}}{0.30} = \frac{1 - 0.223}{0.30} = \mathbf{2.590}$$
$$\ln A(5) = -0.1174 \Rightarrow A(5) = \mathbf{0.889}$$

B(5) is 2.59 - the bond's sensitivity to the short rate over five years - and A(5) works out to 0.889. Both depend only on the parameters and the maturity.

The Price and the Yield

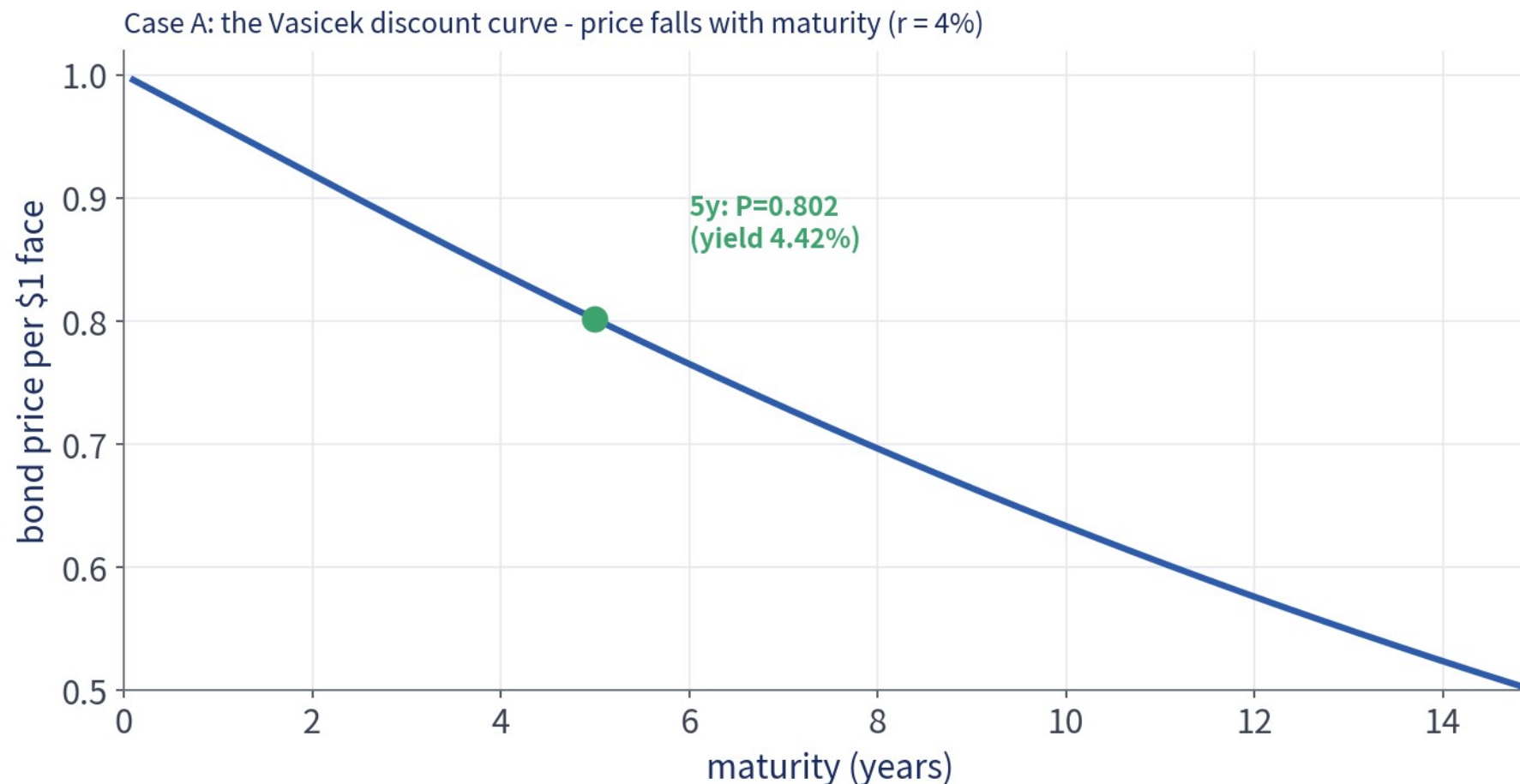
One step from price to curve

$$P(5) = A e^{-Br} = 0.889 e^{-2.590(0.04)} = \mathbf{0.8017}$$
$$y(5) = -\frac{\ln P(5)}{5} = \frac{0.2210}{5} = \mathbf{4.42\%}$$

Plugging in the 4% short rate gives a price of 0.8017 per dollar of face. Taking minus the log over the maturity turns that into a 4.42% yield.

The Discount Curve

Pricing bonds of every maturity at the 4% short rate traces the discount curve: the price falls smoothly as maturity lengthens, because more discounting is applied. The five-year point sits at 0.802 - the bond we priced, yielding 4.42%.



Why Affine Is Powerful

A whole curve from one formula

What affine buys you

- A closed-form price - no simulation needed.
- Every maturity from the same two factors.
- Yields that are linear in the short rate.

Why it matters

- Fast to compute and to calibrate.
- Transparent - you see what drives the price.
- The backbone of tractable rate models.

Build the Curve, Read the Shape

Today's short rate does the tilting

Because the yield is linear in the short rate, one number sets the whole curve. When the short rate sits below the long-run level, the curve slopes up; when it sits above, the curve inverts. We compute yields at four maturities for a low and a high short rate and watch the shape flip

The two scenarios

- Low short rate: 2%, below the long run.
- High short rate: 8%, above the long run.
- Both converge to about 4.78%.

What we'll do by hand

- Compute yields at 1, 3, 5, and 10 years.
- Do it for both short rates.
- Read the slope of each curve.

The Yield, Linear in r

How the curve is built

$$y(T) = \frac{B(T)}{T} r - \frac{\ln A(T)}{T} \quad (\text{yield is linear in today's short rate } r)$$

The short end

- At short maturities, B over T is near one.
- So the yield tracks the short rate closely.
- The near end anchors on today's r .

The long end

- As maturity grows, the yield forgets r .
- It converges to the long-run level.
- About 4.78% for these parameters.

Low Rate: an Upward Curve

Short rate 2%, below the long run

maturity	1y	3y	5y	10y	shape
yield ($r=2\%$)	2.40%	2.99%	3.38%	3.93%	UPWARD

Starting below the long-run level, the yield climbs with maturity toward 4.78% - the normal, upward-sloping curve seen in ordinary times.

Why it rises

- The short rate is pulled up toward theta.
- Longer bonds average in that higher future.
- So longer yields exceed the short rate.

The reading

- An upward curve is the common case.
- It signals expected mean reversion upward.
- No stress in the rate outlook.

High Rate: an Inverted Curve

Short rate 8%, above the long run

maturity	1y	3y	5y	10y	shape
yield ($r=8\%$)	7.59%	6.95%	6.49%	5.83%	INVERTED

Starting above the long-run level, the yield falls with maturity - the inverted curve, historically a recession signal. produced by nothing more than a high short rate.

Why it falls

- The short rate is pulled down toward theta.
- Longer bonds average in that lower future.
- So longer yields fall below the short rate.

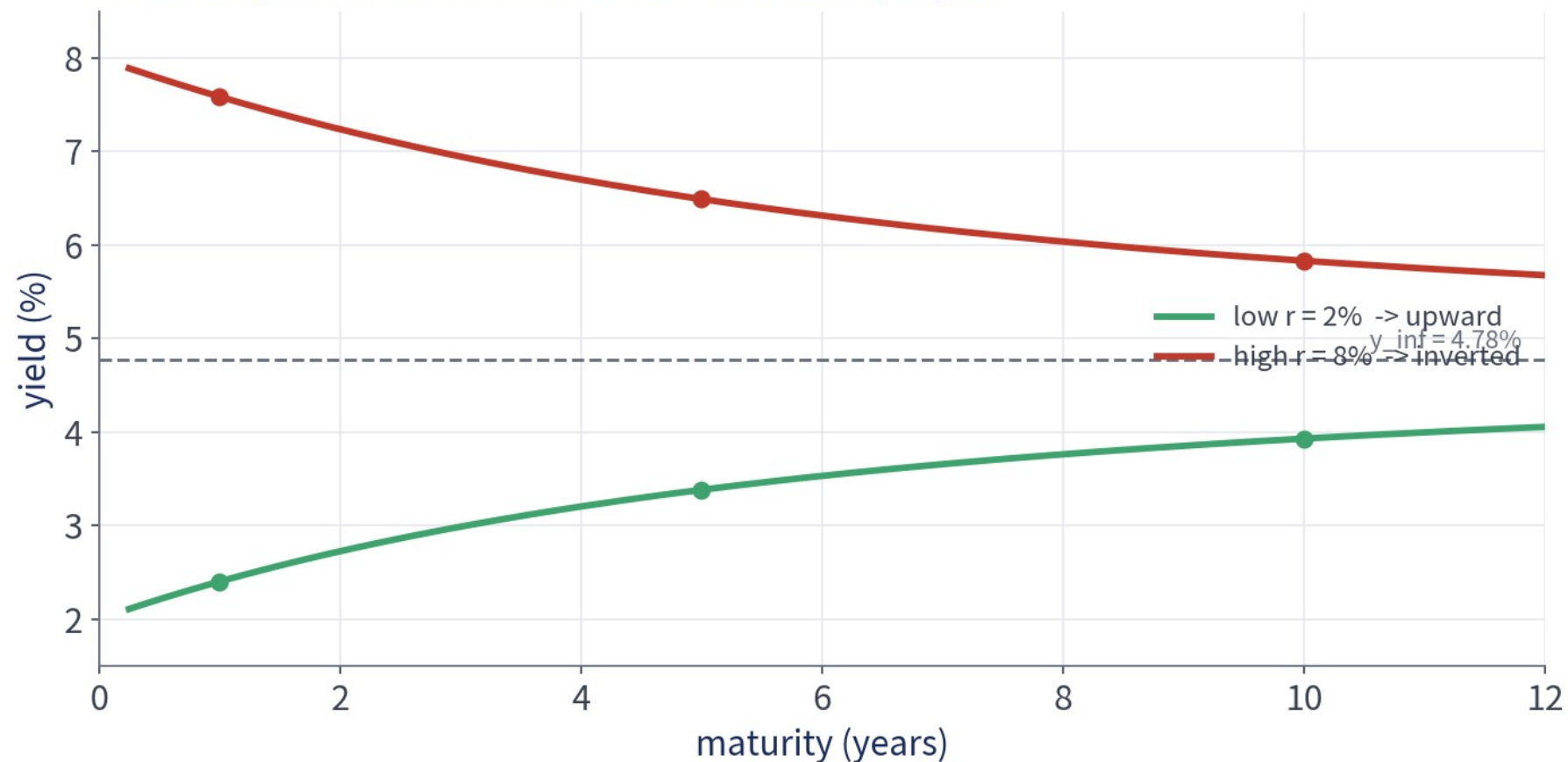
The reading

- An inverted curve flags expected easing.
- The same formula, a different short rate.
- Shape is set entirely by where r starts.

The Short Rate Tilts the Curve

From a low short rate the curve slopes up; from a high one it inverts. Both fan out from today's rate at the short end and converge to the same long-run level of 4.78% at the long end. The entire shape is decided by where the short rate starts

Case B: today's short rate tilts the whole curve - both converge to y_{inf}



The Curve Tells a Story

Shape as a signal

What the shape encodes

- Upward: rates expected to rise toward theta.
- Inverted: rates expected to fall toward theta.
- Humped: a mix in between.

The caveat

- This is the model's mechanism, not a forecast.
- Real curves also carry risk premia.
- But the intuition - r sets the tilt - holds.

One Rate, One Formula, One Curve

The affine engine

The whole term structure comes from one short rate through one affine formula. Price is A times $e^{-B \cdot r}$; yield is that, logged and scaled by maturity. Feed in today's rate and every maturity's yield falls out - and its position relative to the long-run level decides whether the curve rises, inverts, or humps.

Pricing (Case A)

- Affine price from B and A .
- 5-year bond at 0.802, yielding 4.42%.
- Closed form, no simulation.

The curve (Case B)

- Yields linear in the short rate.
- Low r upward, high r inverted.
- All converging to 4.78%.

Reproduce the Price and the Curve

Direct Claude Code to check your arithmetic

GOAL Verify the affine bond price and yield, then plot curves for low and high rates.

[INPUT] Prompt · Claude Code

```
Write and run part2_practice.py that, for Vasicek
kappa=0.30, theta=0.05, sigma=0.02:
CASE A - price a bond:
    * compute B(5), A(5), the 5y price at r=0.04,
      and the yield -ln(P)/5.
CASE B - the curve:
    * print yields at 1,3,5,10y for r=0.02 and r=0.08;
    * label each curve upward or inverted;
    * plot both and mark the long-run yield.
Match every printed number to the slides.
```

Check against the slides

- $B(5)=2.59$, $A(5)=0.889$.
- $P(5)=0.802$, yield 4.42%.
- low r: 2.40 -> 3.93% (up).
- high r: 7.59 -> 5.83% (inverted).

Deliverable: part2_practice.py whose printed numbers match the bond price and the two curves.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What makes the Vasicek price affine?

Price = A times e-to-the-minus-B-times-r

Q2 How do you turn a bond price into a yield?

Minus the log of the price over the maturity

Q3 What decides the curve's shape?

Today's short rate, relative to the long run

All three come from the affine bond price and the yield built from it.

Three Pitfalls to Avoid

Where term structure trips people

Common mistakes

- "The whole curve moves with the short rate equally."
- "An inverted curve is a direct recession forecast."
- "The long yield equals the long-run mean theta."

The reality

- The short end moves most; the long end barely budges.
- It's the model's mechanism - real curves add risk premia.
- The long yield is theta minus a volatility term (4.78%).

Two Calculations to Remember

Part 2 applications in one screen

1

Price a bond with the affine formula

For a 5-year Vasicek bond, $B(5) = 2.59$ and $A(5) = 0.889$; at a 4% short rate the price is 0.8017 per dollar of face, and minus its log over five years gives a 4.42% yield. A closed form, no simulation.

2

Build the curve from the short rate

Because yields are linear in r , a low 2% rate produced an upward curve (2.40% to 3.93%) and a high 8% rate an inverted one (7.59% to 5.83%), both converging to the 4.78% long-run level. The short rate tilts everything.

3

One rate, one formula, one curve

The affine structure turns a single short rate into a price and a yield for every maturity. Where that rate sits relative to the long-run level decides whether the curve rises, inverts, or humps.

Next — Part 3 practice: discretizing and calibrating these models to real market data.

★ WEEK 12 · PART 3 of 3

Discretization, Calibration, and Pricing

Making the Models Run and Match Reality

Euler-Maruyama · Milstein · calibration · pricing by simulation

To use a short-rate model you must run it on a computer and fit it to data. We discretize the SDEs, calibrate the parameters, and price a bond end-to-end by simulation — checked against the closed form.

3

PART 3 · COMPUTATION

From SDE to a Priced Bond

Discretizing the SDEs with Euler-Maruyama and Milstein, the pitfalls to avoid, calibrating to market data, and pricing a bond by simulation — the full computational workflow.

From Formulas to Working Code

Parts 1-2 gave the theory; Part 3 runs it

1

We have models and formulas

Vasicek and CIR, and closed-form bond prices and yields. But real work needs to simulate, fit, and price numerically.

2

SDEs must be discretized

A computer can't take dt to zero. We approximate the SDE with finite steps — Euler-Maruyama, and the finer Milstein.

3

Then calibrate and price

Fit the parameters to data, simulate paths, and average discounted payoffs into a price — validated against the closed form.

Part 3 in one line: discretize the SDE, calibrate to data, and price a bond by simulation.

Why Discretize?

Computers can't take dt to zero

An SDE is written in infinitesimal time, but a computer works in finite steps. Discretization replaces the continuous dt with a small but finite step, turning the SDE into a recipe a machine can actually run.

The gap

- The SDE lives in continuous time, with dt going to zero.
- A computer must take steps of some finite size.
- We need a rule to advance the state one step.

The pay-off

- Simulate any model, even without a closed form.
- Generate paths for pricing and risk.
- The foundation for Monte Carlo pricing.

Euler-Maruyama

The simplest way to step an SDE

$$\text{Euler-Maruyama: } X_{t+\Delta t} = X_t + a(X_t) \Delta t + b(X_t) \sqrt{\Delta t} Z, \quad Z \sim \mathcal{N}(0, 1)$$

Reading it

- Advance the drift by a term proportional to dt .
- Add a random shock scaled by the root of dt .
- The shock is a standard normal draw.

Why root- dt

- Brownian increments have variance dt .
- So their standard deviation scales as root- dt .
- That is the hallmark of a stochastic step.

Reading the Euler Step

Drift by dt , noise by $\sqrt{dt}$

The drift part

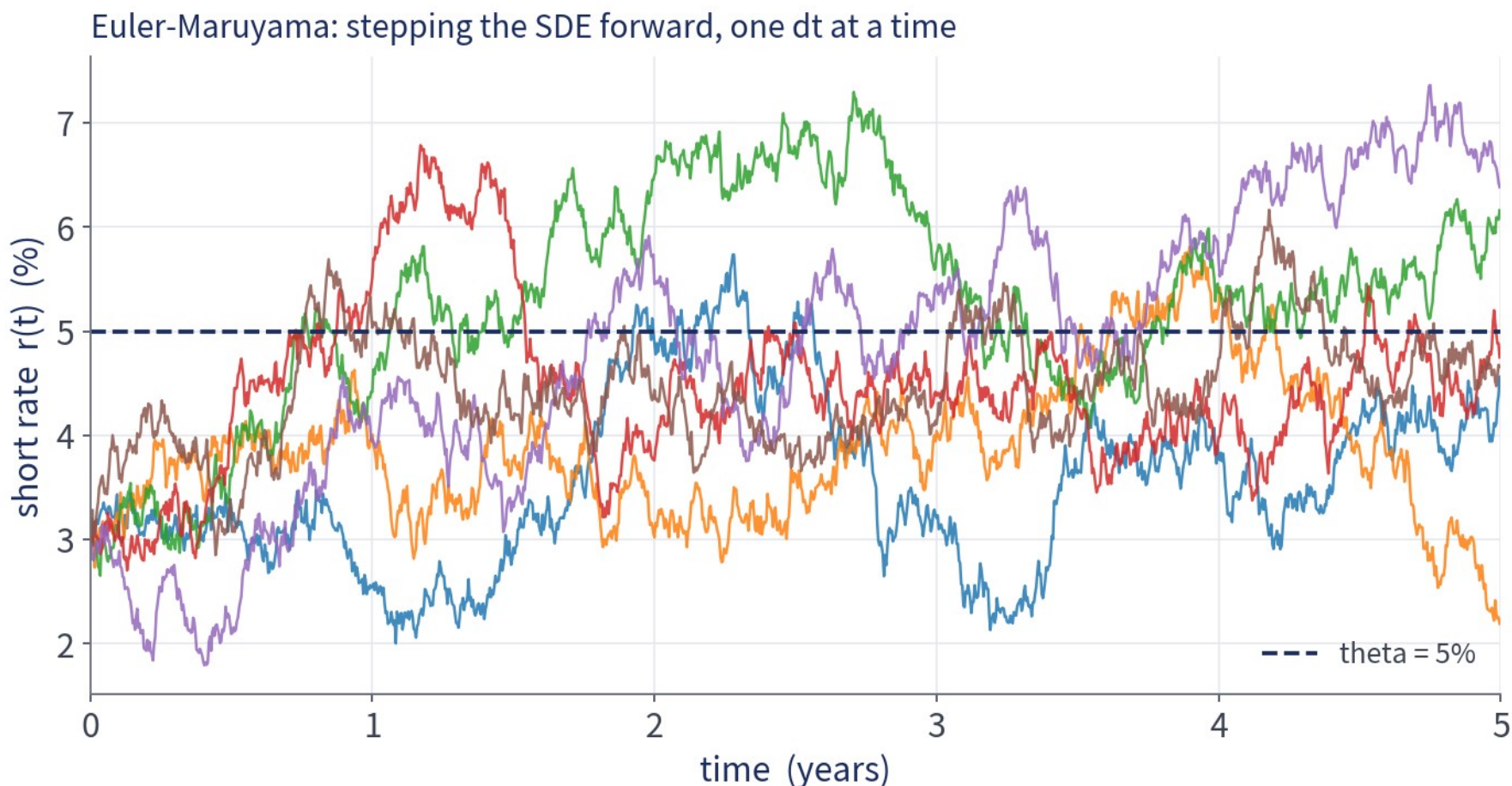
- Deterministic: the trend times the step size.
- For Vasicek, it pulls the rate toward θ .
- Scales linearly with dt .

The diffusion part

- Random: the diffusion times a normal draw.
- Scales with the square root of dt , not dt .
- This is what makes the path jagged.

Euler-Maruyama in Action

Stepping the Vasicek SDE forward one dt at a time produces sample rate paths that revert toward θ . This is the raw material for simulation-based pricing and risk.



Euler's Pitfalls

Discretization error and negative rates

Euler is simple but imperfect. Too large a step introduces discretization error, and for CIR the scheme can push the rate below zero — where the square root is undefined. Both need care.

Discretization error

- A finite step only approximates the true SDE.
- Larger steps mean larger error.
- Shrinking dt improves accuracy — at a cost.

The CIR zero problem

- A big negative shock can drive CIR's rate below zero.
- Then the square root of the rate is undefined.
- Naive Euler breaks exactly where CIR should be safe.

The Negative-Value Fix for CIR

Keeping the square root well-defined

Simple fixes

- Full truncation: use the positive part of the rate inside the root.
- Reflection: bounce a negative value back above zero.
- Floor the rate at zero after each step.

The exact scheme

- CIR has a known transition (non-central chi-squared).
- Sampling from it avoids the problem entirely.
- More work, but exact and always non-negative.

Milstein: One Correction Term

More accuracy for level-dependent diffusion

$$\text{Milstein: } + \frac{1}{2} b(X_t) b'(X_t) [(\sqrt{\Delta t} Z)^2 - \Delta t] \leftarrow \text{extra correction}$$

What it adds

- The Euler step, plus a second-order correction.
- The correction uses the diffusion's derivative.
- It matters most when diffusion depends on the state.

The benefit

- It captures curvature Euler misses.
- It converges faster as the step shrinks.
- For constant-diffusion Vasicek, it reduces to Euler.

Euler vs Milstein

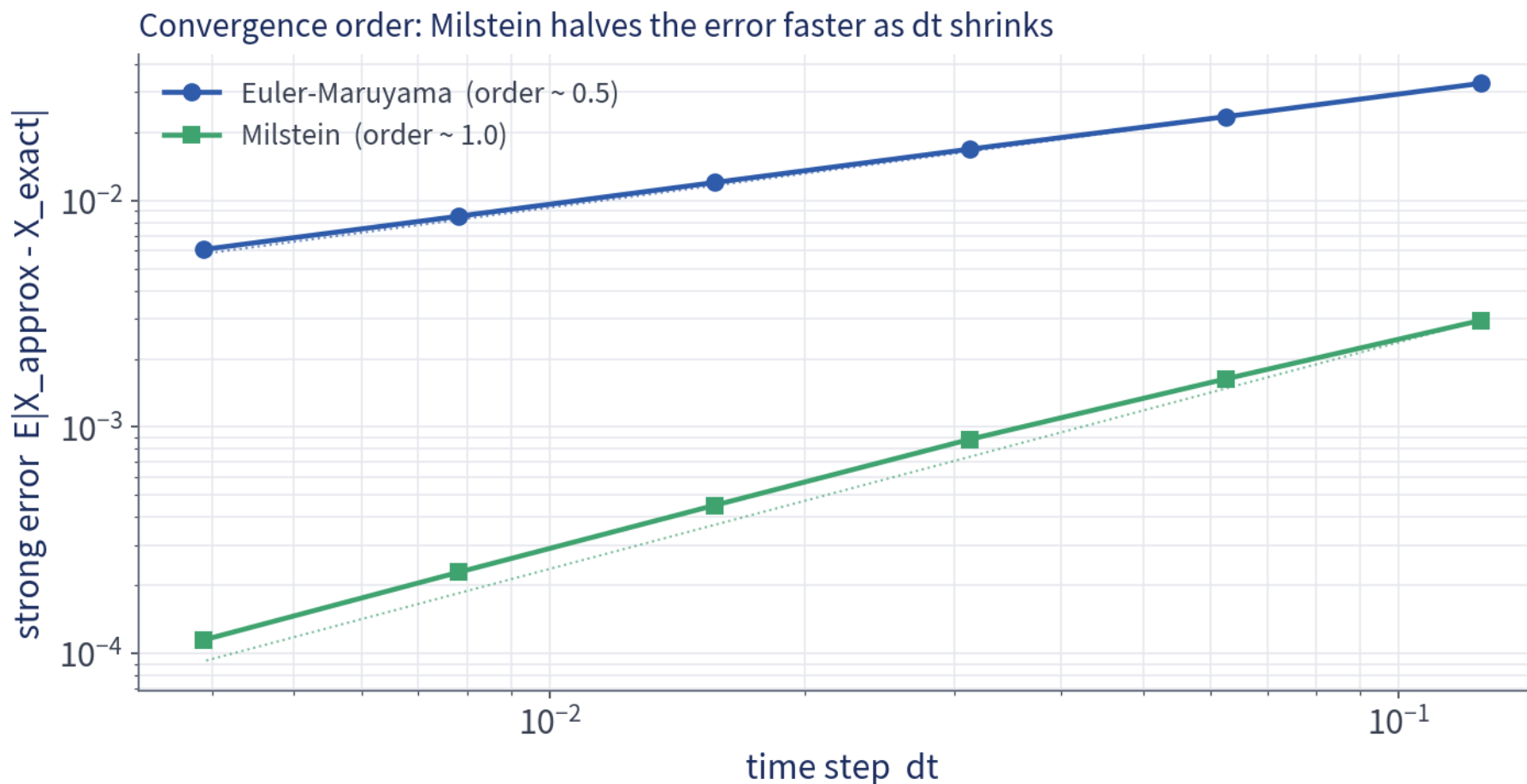
Simplicity against accuracy

Aspect	Euler-Maruyama	Milstein
Update	drift dt + diffusion $\sqrt{dt}$ Z	plus a correction term
Strong order	about 0.5	about 1.0
Needs	only drift and diffusion	also the diffusion's derivative
Cost per step	cheapest	slightly more
When to use	quick default	when accuracy matters

Milstein halves the error faster as dt shrinks — at the price of one extra term.

Convergence Order

Strong error versus step size, on a log-log scale. Euler's error falls with slope about one-half; Milstein's falls with slope about one — it reaches a given accuracy with far coarser steps.



Calibrating to Market Data

Making the model match reality

A model is only useful once its parameters reflect reality. Calibration fits κ , θ , and σ so the model agrees with data — either the historical behavior of rates, or today's observed prices.

The goal

- Choose parameters so the model matches the world.
- Either match how rates have MOVED, or how bonds are PRICED.
- Different data answer different questions.

Two regimes

- Time-series fit: estimate from historical rates.
- Cross-sectional fit: match today's yield curve.
- Sometimes both, for a full picture.

Two Sources of Data

History versus prices

Historical rates

- A time series of the short rate over time.
- Estimate the real-world dynamics: reversion and noise.
- Uses the AR(1) bridge from Week 11.

Market prices

- Today's observed bond prices or yields.
- Fit parameters so the model curve matches them.
- Gives the risk-neutral parameters for pricing.

Estimating From History

The AR(1) bridge, one more time

from history: regress r_{t+1} on $r_t \Rightarrow \phi \Rightarrow \kappa = -\frac{\ln \phi}{\Delta t}, \theta, \sigma$

The recipe

- Regress the rate on its own one-step lag.
- The slope gives the persistence ϕ .
- Convert ϕ into κ , θ , and σ .

Why it works

- Discretized Vasicek is exactly an AR(1).
- So the same estimator from Week 11 applies.
- A one-line regression yields all three parameters.

Calibrating to the Curve

Matching today's observed yields

The procedure

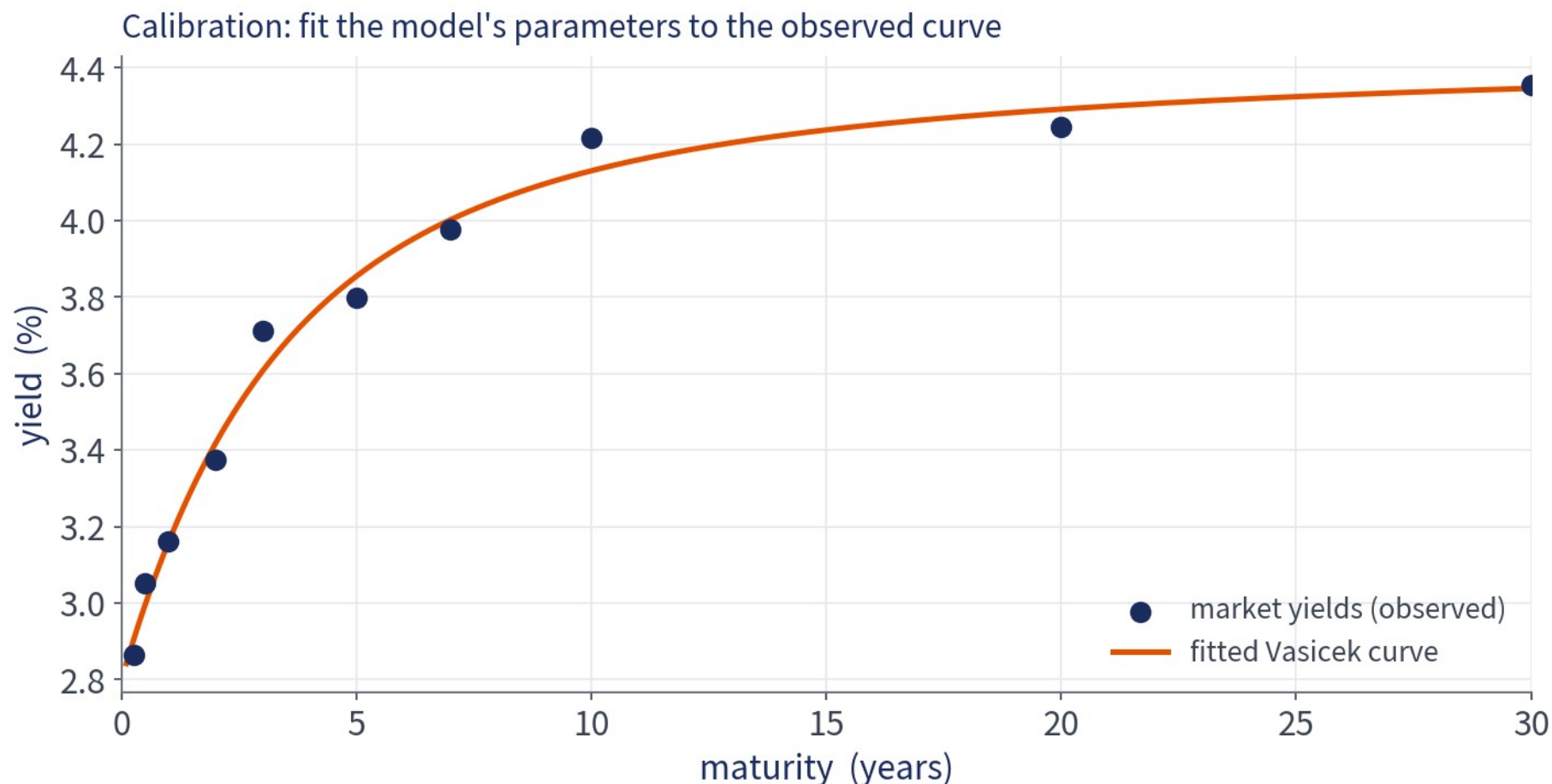
- Take the observed yields across maturities.
- Search for parameters whose model curve fits them.
- Minimize the gap between model and market.

Why it differs from history

- It targets prices, not past dynamics.
- It gives the parameters used for pricing.
- The two fits can, and often do, differ.

Calibration to the Yield Curve

Choosing the parameters so the model's yield curve passes through the observed market yields. A good fit means the model reproduces today's prices before you trust it for anything else.



Pricing a Bond by Simulation

The end-to-end approach

With a calibrated model you can price by simulation: generate many short-rate paths, discount along each, and average. For Vasicek and CIR you can check the answer against the closed form — then trust the engine on harder products.

Why simulate at all

- Many real products have no closed-form price.
- Simulation handles them uniformly.
- The bond is the simplest case to validate on.

The validation

- Price the bond both ways: simulation and formula.
- They should agree as paths increase.
- Agreement builds trust in the simulator.

The Simulation Workflow

Simulate, discount, average

1

Simulate paths

Step the calibrated SDE forward with Euler or Milstein to generate many short-rate paths to the bond's maturity.

2

Discount each path

Along each path, accumulate the rate over time and form its discount factor, the exponential of minus that accumulated rate.

3

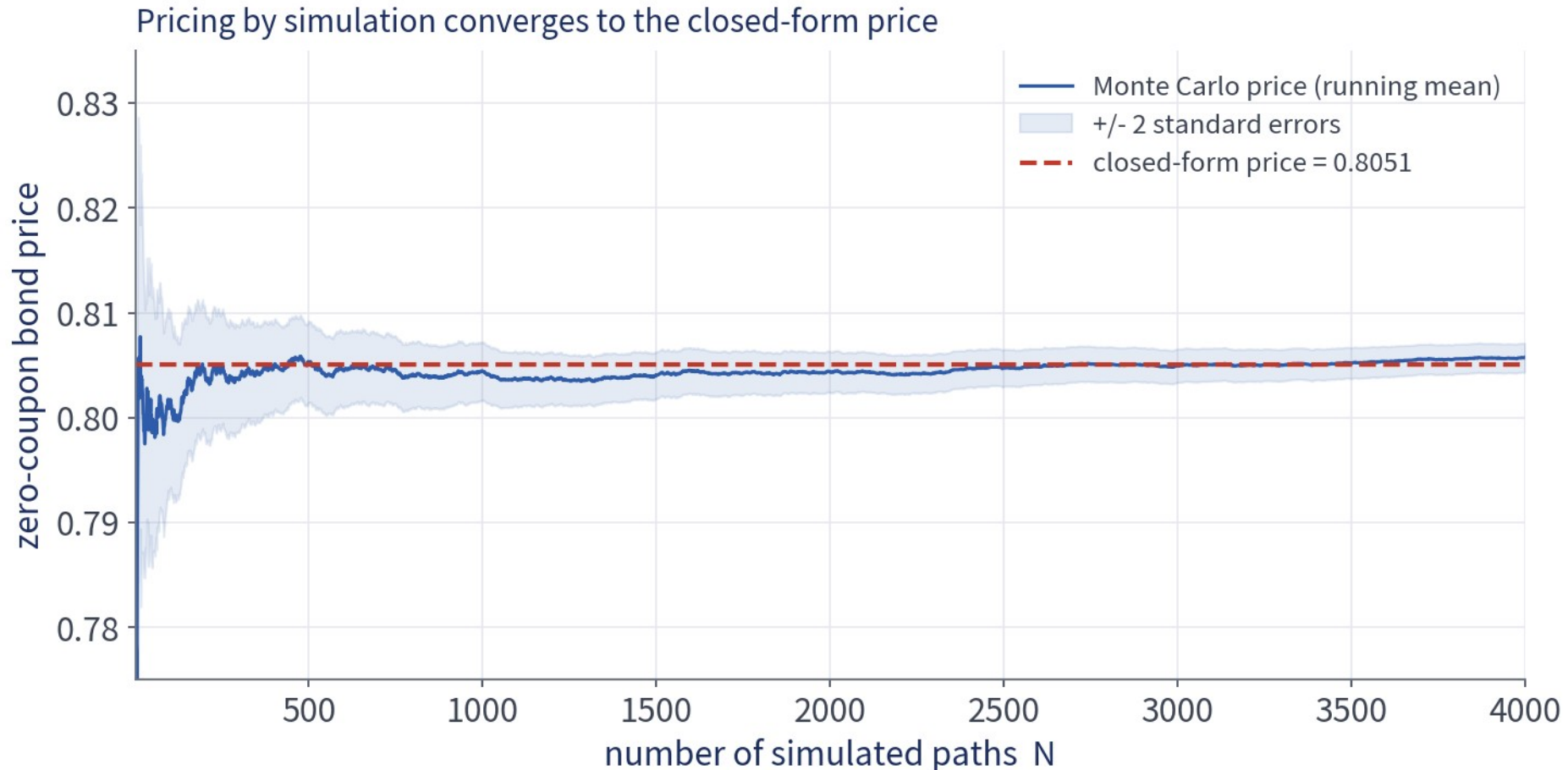
Average across paths

The bond price is the mean discount factor over all paths. More paths shrink the standard error, tightening the estimate.

The price is just an average of discounted payoffs — the pricing expectation, made numerical.

Simulation Converges to the Closed Form

The Monte Carlo bond price, as a running average over simulated paths, converges to the analytic Vasicek price. The band of two standard errors narrows as the number of paths grows.

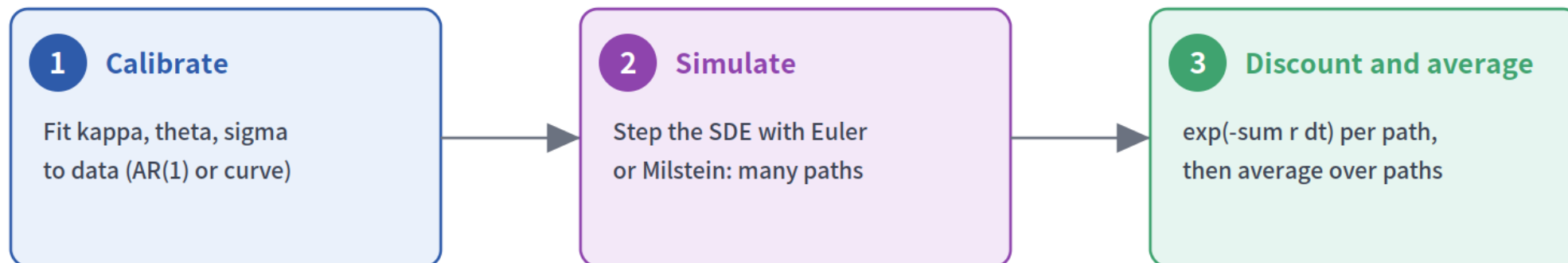


The End-to-End Pricing Pipeline

Calibrate the parameters, simulate paths by Euler or Milstein, then discount and average into a price. Always cross-check against the closed form before trusting the engine on products that lack one.

The End-to-End Pricing Pipeline

Three stages turn raw data into a bond price: calibrate, simulate, discount-and-average.



Always cross-check against the closed form

For Vasicek and CIR you can price a zero-coupon bond analytically. Use that to VALIDATE the simulator: the Monte Carlo price should converge to the closed-form price as the number of paths grows.

Simulation shines for products with no closed form - once the engine is trusted.

Simulate or Use the Formula?

Match the tool to the task

Use the closed form when

- You are pricing a plain bond under Vasicek or CIR.
- Speed and exactness matter.
- A formula exists — there's no reason to simulate.

Simulate when

- The product has no closed-form price.
- Payoffs are path-dependent or exotic.
- You need a flexible, general-purpose engine.

Calibrate and Price a Zero-Coupon Bond

Direct Claude Code through the full pipeline

GOAL Calibrate a short-rate model, price a bond by simulation, and check it against the closed form.

[INPUT] Prompt · Claude Code

Write and run `price_bond.py` that:

1. estimates Vasicek κ , θ , σ from a short-rate series via the AR(1) regression;
 2. simulates many short-rate paths with Euler and discounts each: $\exp(-\sum r \cdot dt)$;
 3. prices a zero-coupon bond as the average discount, with a standard error;
 4. compares to the closed-form Vasicek price and plots convergence as the path count grows.
- Confirm the simulated price converges to the closed form within the error band.

How to work with it

- AR(1) gives the parameters.
- Discount = $\exp(-\sum r \cdot dt)$.
- More paths shrink the error.
- It should match the closed form.

Deliverable: `price_bond.py` + a calibrated model, an MC bond price, and a closed-form check.

Three Questions Before You Finish

If you can answer these, Week 12 has landed

Q1	In the Euler step, the random shock scales with what power of dt ?	The square root
Q2	What does Milstein add over Euler?	A second-order correction
Q3	How do you validate a simulated bond price?	Against the closed form

All three come from the Euler step, the Milstein correction, and the closed-form check.

Three Pitfalls to Avoid

Where the numerics go wrong

Common mistakes

- "A big time step is fine — it's roughly right."
- "Naive Euler is safe for CIR."
- "A simulated price needs no validation."

The reality

- Large steps add real error; check sensitivity to dt .
- Euler can push CIR negative — truncate or use the exact scheme.
- Always cross-check against the closed form when one exists.

Week 12 in Three Ideas

Pulling the whole week together

1

Rates revert — Vasicek and CIR

Parts 1-2: the short rate is a mean-reverting SDE. Vasicek is OU for the rate; CIR keeps it positive with a square-root diffusion. Both are affine, giving closed-form bond prices and whole yield curves.

2

Discretize to compute

Euler-Maruyama steps the SDE; Milstein adds a correction for faster convergence. Mind the pitfalls — discretization error and CIR's zero boundary.

3

Calibrate, simulate, and validate

Fit the parameters to history or the curve, price a bond by simulating and averaging discounted payoffs, and check the result against the closed form.

Next — Week 13: Bayesian inference, treating those calibrated parameters as distributions, not points.

★ WEEK 12 · PART 3 · PRACTICE

Discretize and Calibrate, Worked by Hand

Step an SDE, Fit It, and Price by Simulation

Euler-Maruyama, Milstein, and a calibration workflow

Part 3's numerical toolkit, with the arithmetic shown. We take an Euler step by hand and see the negative-rate pitfall Milstein repairs, then calibrate Vasicek from history and price a bond by simulation - checking it against the closed form.

3P

PART 3 · APPLICATIONS

From Formula to Simulation

Case A: an Euler-Maruyama step, the negative-rate pitfall it can hit, and the Milstein correction that repairs it. Case B: calibrate Vasicek via the AR(1) bridge and price a bond by simulation against the closed form. Then a Claude Code lab.

What Part 3 Told Us

Now we step and fit - with numbers

1

Euler-Maruyama steps an SDE

Drift scaled by dt , noise by $\sqrt{dt}$. Simple and general - but it makes discretization error and can break positivity.

2

Milstein adds one correction

For a level-dependent diffusion, an extra term sharpens accuracy and helps keep rates positive.

3

Calibrate, then simulate to price

Fit the parameters from history or the curve, then simulate, discount, and average to price - checked against the closed form.

Everything here is a calculator exercise: one Euler step, one calibration, one simulated price.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from an SDE or a data series.
- Take a step, calibrate, or simulate by hand.
- Then read the numerical lesson.

What you should watch

- How a big shock can push Euler below zero.
- How Milstein repairs it.
- How the simulated price matches the formula.

An Euler Step, and Its Pitfall

Where discretization breaks positivity

To simulate a rate model, we replace the continuous SDE with discrete steps. Euler-Maruyama is the simplest: move by the drift times dt and the noise times $\sqrt{dt}$. We take one step of CIR by hand, then see how a large negative shock can push the discretized rate below zero - and how Milstein repairs it

The setup

- CIR: $\kappa = 0.30$, $\theta = 5\%$, $\sigma = 0.20$.
- Monthly step, starting rate 1%.
- Near the floor, where trouble lives.

What we'll do by hand

- Compute the drift and noise pieces.
- Step with a normal and a large shock.
- Apply the Milstein correction.

The Euler-Maruyama Step

Drift by dt, noise by root-dt

$$r_{t+\Delta t} = r_t + \underbrace{\kappa(\theta - r_t)\Delta t}_{\text{drift} \times \Delta t} + \underbrace{\sigma\sqrt{r_t}\sqrt{\Delta t}z}_{\text{noise} \times \sqrt{\Delta t}} \quad (z \sim N(0, 1))$$

The drift piece

- The reversion, scaled by the time step.
- It's deterministic and usually small.
- It nudges the rate toward theta.

The noise piece

- The diffusion, scaled by root-dt.
- A random normal shock each step.
- For CIR it also scales with root-r.

One Euler Step

The drift and noise pieces

$$\begin{aligned}\text{drift} &= 0.30(0.05 - 0.01)\left(\frac{1}{12}\right) = 0.0010 \\ \text{noise coef} &= 0.20\sqrt{0.01}\sqrt{\frac{1}{12}} = 0.00577\end{aligned}$$

At a 1% rate, the drift adds a tiny 0.0010, while the noise coefficient is 0.00577 per unit shock - small, but a large multiple of it can dominate the step.

The Pitfall, and the Fix

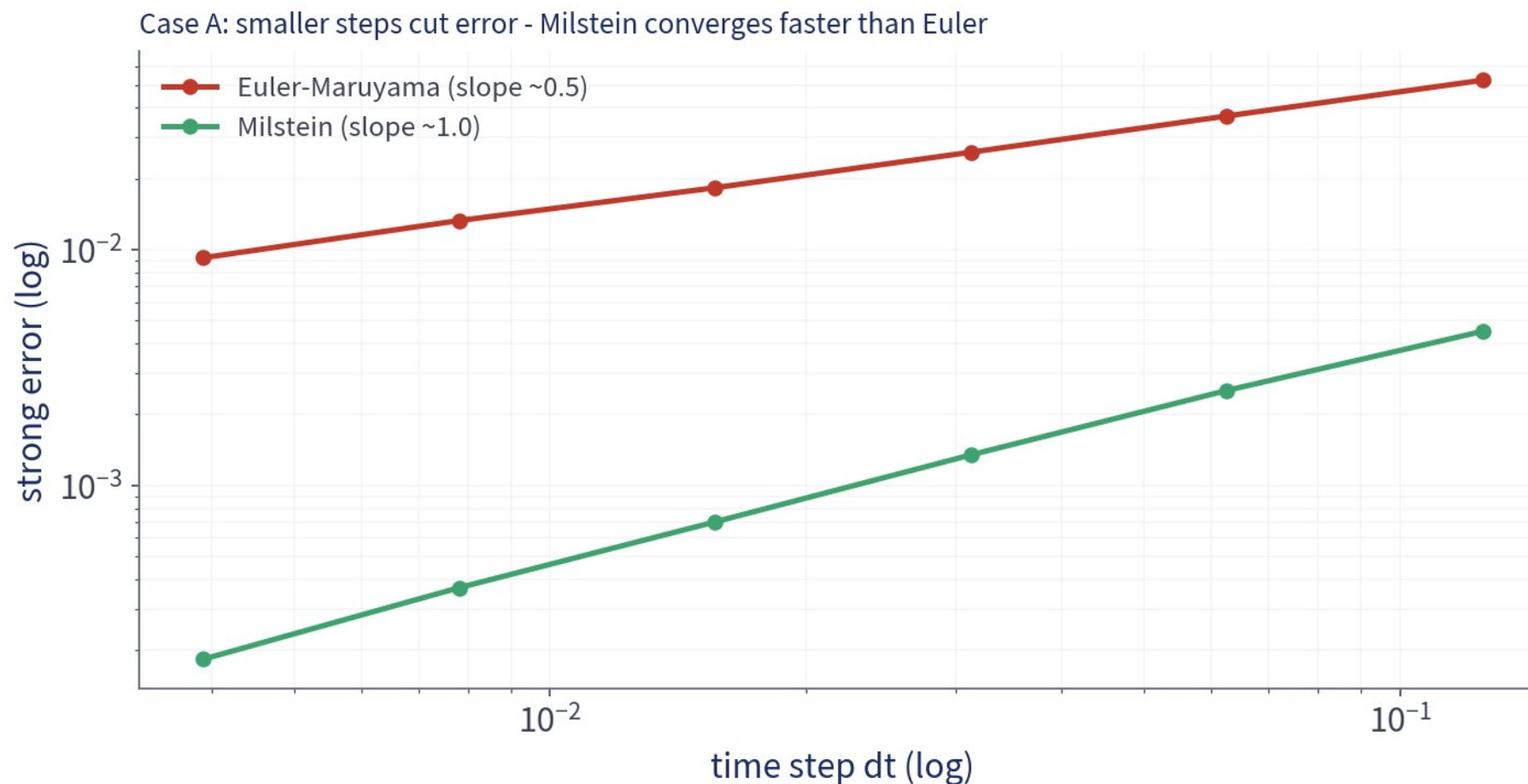
A big shock breaks the floor

$$z = -3 : r_{t+\Delta t} = 0.01 + 0.0010 - 3(0.00577) = -\mathbf{0.0063} \text{ (negative!)}$$
$$\text{Milstein} + \frac{\sigma^2}{4}(z^2 - 1)\Delta t = +0.0067 \Rightarrow r_{t+\Delta t} = +0.0004 \text{ (repaired)}$$

A minus-three shock drives the Euler step to -0.0063 - a negative rate that continuous CIR can never reach. The Milstein correction adds 0.0067 and repairs it back above zero.

Smaller Steps, Less Error

On a log-log plot, both schemes' error shrinks as the step size falls - but Milstein falls twice as fast (slope near 1) as Euler (slope near 0.5). For the same accuracy, Milstein lets you take far larger, cheaper steps.



Choose the Scheme Deliberately

Accuracy and positivity both matter

Euler's weaknesses

- Error shrinks only with root-dt.
- A big shock can push rates negative.
- Fine for rough work, risky near a floor.

When to reach for Milstein

- Level-dependent diffusion, like CIR's root-r.
- Faster convergence, better positivity.
- One extra term, well worth it near zero.

Calibrate, Then Price by Simulation

Fit the model, then trust it

A model is only useful once it's fit to data. We calibrate Vasicek from a monthly rate history through the AR(1) bridge - one more time - recovering the reversion speed, long-run level, and volatility. Then we price a five-year bond by simulation and check it against the affine closed form from Part 2

The history

- Monthly AR(1): coefficient 0.96.
- Intercept 0.002, residual sd 0.005.
- Sampled monthly - dt is one twelfth.

What we'll do by hand

- Map to kappa, theta, and sigma.
- Price a 5-year bond by simulation.
- Compare to the closed form.

Calibration via the AR(1) Bridge

Three parameters from a regression

$$\kappa = -\frac{\ln \phi}{\Delta t}, \quad \theta = \frac{c}{1-\phi}, \quad \sigma = \sigma_\varepsilon \sqrt{\frac{2\kappa}{1-\phi^2}} \quad (\text{AR}(1) \rightarrow \text{Vasicek})$$

The mapping

- kappa from the coefficient, as in Week 11.
- theta from the intercept over one-minus-phi.
- sigma from the residual sd, annualized.

The principle

- Estimate discretely with a simple regression.
- Interpret the result in continuous time.
- One line of code, three model parameters.

The Fitted Parameters

kappa, theta, sigma from history

$$\begin{aligned}\kappa &= -\ln(0.96) \times 12 = \mathbf{0.49}/\text{yr} \quad (\text{half-life } 1.4 \text{ yr}) \\ \theta &= \frac{0.002}{0.04} = \mathbf{5.0\%}, \quad \sigma = 0.005\sqrt{12.5} = \mathbf{1.77\%}\end{aligned}$$

The AR(1) fit gives a reversion speed of 0.49 per year - a 1.4-year half-life - a long-run level of 5%, and a volatility of 1.77%. A full Vasicek model, from one regression.

Price by Simulation

Simulate, discount, average

$$P = \mathbb{E} \left[e^{-\int_0^T r dt} \right] : \quad \text{Monte Carlo } 0.8099 \pm 0.0002 \approx \text{closed form } 0.8096$$

The workflow

- Simulate many short-rate paths with Euler.
- Discount each by its own integrated rate.
- Average the discount factors.

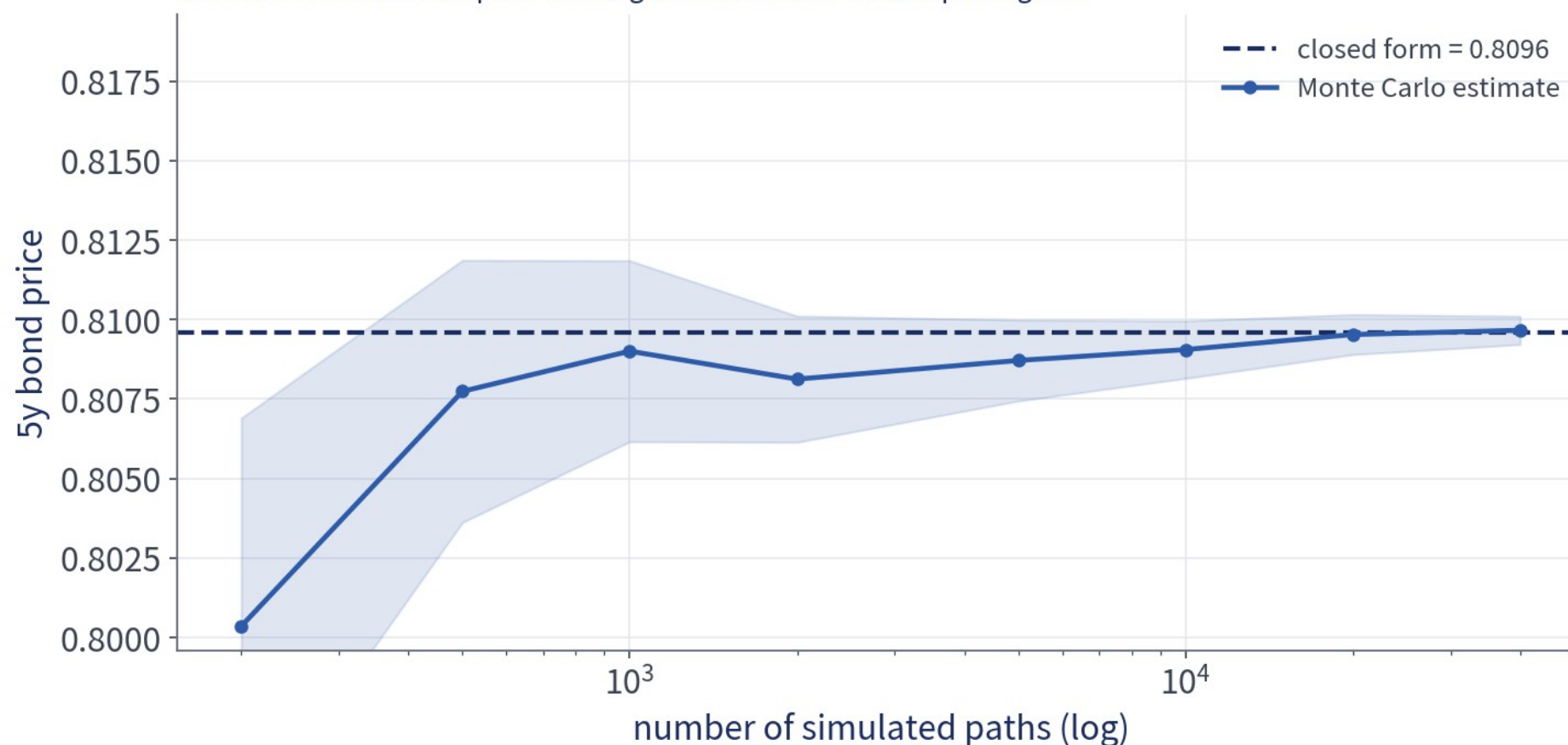
The check

- The Monte Carlo price is 0.8099.
- The closed form is 0.8096.
- They agree - the simulation is validated.

Simulation Meets the Formula

With only a few hundred paths the Monte Carlo price is noisy, but as the path count grows it converges tightly onto the closed-form value. The shrinking band is the shrinking standard error - and its landing on the dashed line confirms the simulation is correct

Case B: the Monte Carlo price converges to the closed form as paths grow



Why Both Data and Simulation

Fit to reality, then compute anything

Two sources of data

- History calibrates the real-world dynamics.
- Prices calibrate to today's observed curve.
- Which you use depends on the question.

Why simulate at all

- The closed form only exists for simple models.
- Simulation prices anything - once validated.
- Check against the formula where you can.

The Full Numerical Workflow

Discretize, calibrate, simulate, g
Part 3 turns the theory into computation. Discretize the SDE to step it - choosing Euler or Milstein for the accuracy and positivity you need. Calibrate the parameters from history or prices. Then simulate, discount, and average to price - validating against the closed form wherever one exists. That loop is how rate models

Discretize (Case A)

- Euler for simplicity, Milstein for accuracy.
- Watch the negative-rate pitfall.
- Smaller steps, less error.

Calibrate and price (Case B)

- AR(1) bridge recovers the parameters.
- Simulate, discount, average to price.
- Validate against the closed form.

Reproduce the Step and the Price

Direct Claude Code to check your arithmetic

GOAL Verify the Euler step and Milstein fix, calibrate Vasicek, and price by simulation.

[INPUT] Prompt · Claude Code

Write and run part3_practice.py that:

CASE A - Euler:

- * for CIR ($k=0.30$, $\theta=0.05$, $\sigma=0.20$, $dt=1/12$, $r=0.01$), print the Euler step for $z=+1$ and $z=-3$;
- * add the Milstein term and show it repairs $z=-3$.

CASE B - calibrate + price:

- * from AR(1) $\phi=0.96$, $c=0.002$, $se=0.005$, print κ , θ , σ ;
- * price a 5y bond by simulation and compare to the closed form.

Match every printed number to the slides.

Check against the slides

- Euler $z=-3$: -0.0063 (negative).
- Milstein repairs to +0.0004.
- κ 0.49, θ 5%, σ 1.77%.
- MC 0.8099 vs closed 0.8096.

Deliverable: part3_practice.py whose printed numbers match the Euler step, calibration, and price.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How does the noise scale in an Euler step?

With root-dt, not dt

Q2 Why can Euler give a negative rate?

**A big shock
overshoots the zero
floor**

Q3 How do you price a bond by simulation?

**Simulate, discount
each path, average**

All three come from the Euler step, the negative-rate pitfall, and the simulation workflow.

Three Pitfalls to Avoid

Where discretization trips people

Common mistakes

- "Scale the noise by dt like the drift."
- "Euler is always accurate enough."
- "Trust a simulated price with no check."

The reality

- Noise scales with $\sqrt{dt}$ - it dominates small steps.
- Near a floor, Euler breaks positivity; use Milstein.
- Validate against the closed form whenever one exists.

Two Calculations to Remember

Part 3 applications in one screen

1

Step carefully, and pick the scheme

An Euler step of CIR at 1% had a drift of 0.0010 and a noise coefficient of 0.00577; a minus-three shock pushed it to -0.0063, a negative rate, which the Milstein correction of 0.0067 repaired. Smaller steps and Milstein cut the error.

2

Calibrate, then price by simulation

The AR(1) bridge recovered $\kappa = 0.49$, $\theta = 5\%$, and $\sigma = 1.77\%$ from a monthly history. Simulating, discounting, and averaging gave a 5-year price of 0.8099, matching the closed form's 0.8096 - the simulation validated.

3

The full numerical workflow

Discretize to step, calibrate to fit, simulate to price, and check against the formula. That loop takes a rate model from the page to a number you can trade on.

Next — Week 13: the course turns from continuous-time models toward machine learning for financial time series.